COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4

Module 4
Reliable Data Transfer: ARQ Protocols

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RECALL · MODULE 3

A quick recap

- Datagrams can get **corrupted** as they travel through the network.
- **Error detection codes** — parity check, Internet checksum, CRC — let the receiver detect whether a packet has been corrupted.
- **Forward Error Correction (FEC)** goes further: using error-correcting codes, the receiver can not only detect errors but also **correct** them without asking for a retransmission.

THE BIGGER PICTURE

What error correction cannot solve

Error correction alone cannot handle every failure. Two cases in particular:

CASE 1 — CORRUPTED BEYOND REPAIR

The packet arrives corrupted, and the corruption is **beyond correction**, so the receiver discards it. The sender receives **no signal** — as far as it knows, the packet was delivered.

CASE 2 — NEVER ARRIVES

The packet is **lost in transit** and never arrives at all. There is nothing to correct and nothing to detect — and again, **the sender has no idea**.

Error correction alone is not enough. We need a **concrete protocol between the sender and the receiver** to coordinate and handle both cases.

MODULE 4 · CONCEPT 1 OF 4

Reliable Data Transfer over an Unreliable Channel

Reliable Data Transfer and Automatic Repeat reQuest

1 · Reliable data transfer | 2 · Stop-and-wait RDT | 3 · Go-back-N | 4 · Selective repeat

THIS MODULE

Reliable Data Transfer (RDT)

Reliable Data Transfer (RDT) is the goal: deliver data correctly, completely, and in order to the layer above (typically the application) — no matter what the underlying channel does to it.

For example, suppose the underlying channel is **IP**. IP is only **best-effort**: packets might be lost, might arrive corrupted, or might arrive out of order.

If we run an RDT protocol on top of IP, it compensates for all of those risks and still provides correct, complete, in-order delivery.

MODULE 4

Outline

We build up **reliable data transfer (RDT)** protocols one channel assumption at a time:

- **rdt 1.0** — over a perfectly reliable channel
- **rdt 2.0** — over a channel with **bit errors**
- **rdt 3.0** — over a **lossy** channel with bit errors

ARQ

ARQ — Automatic Repeat reQuest

Most RDT protocols are built on a core idea: if the sender does not hear back, it **retransmits**. Protocols following this pattern are called **ARQ** (Automatic Repeat reQuest) protocols.

By the end of this module, we will learn three ARQ protocols, in increasing sophistication:

- **Stop-and-wait** ARQ — the basic version, the ARQ underlying **rdt 3.0**.
- Building on it, a **pipelined** and more efficient version: **go-back-N** ARQ.
- **Selective repeat** ARQ — pipelined, and more efficient still.

MODULE 4 · CONCEPT 2 OF 4

Stop-and-wait RDT

Building the protocol up from rdt 1.0 to rdt 3.0

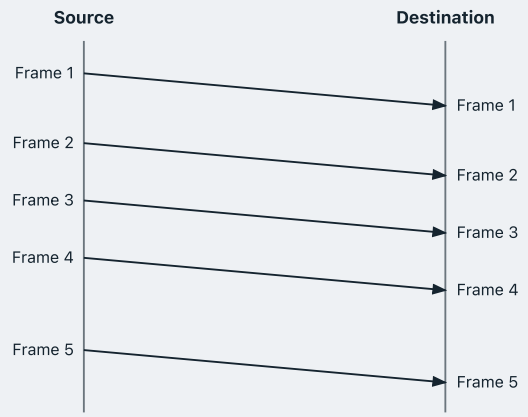
1 · Reliable data transfer | **2 · Stop-and-wait RDT** | 3 · Go-back-N | 4 · Selective repeat

rdt 1.0 — a perfectly reliable channel

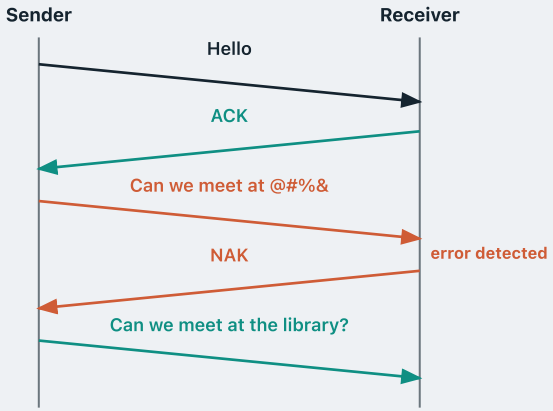
Start with the easiest case — a **perfectly reliable channel**:

- no **bit errors**
- no **out-of-order** frames
- no **frame loss**

> Nothing can go wrong, so there is nothing for the protocol to do: the sender just sends, and the receiver just delivers what arrives.



rdt 2.0 — a channel with only bit errors



Bits can now be flipped in transit.

Here's one idea: the receiver runs an **error detection code** — a checksum or CRC — exactly what we learned in Module 3.

If it arrives intact, the receiver sends a **positive acknowledgement (ACK)**.

If an error is spotted, the receiver sends a **negative acknowledgement (NAK)**.

On receiving a NAK, the sender **retransmits**.

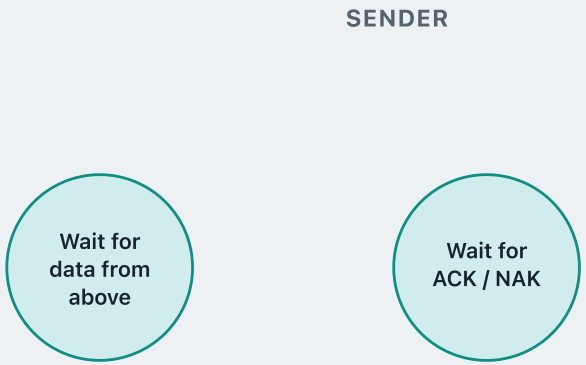
RDT 2.0

The protocol as a state machine

- We describe what the sender does and what the receiver does with one **finite-state machine** each.
- A finite-state machine sits in one of several **states**, with arrows for the **transitions** between them — and each arrow is labelled with the event that triggers that transition.

RDT 2.0

The protocol as a state machine



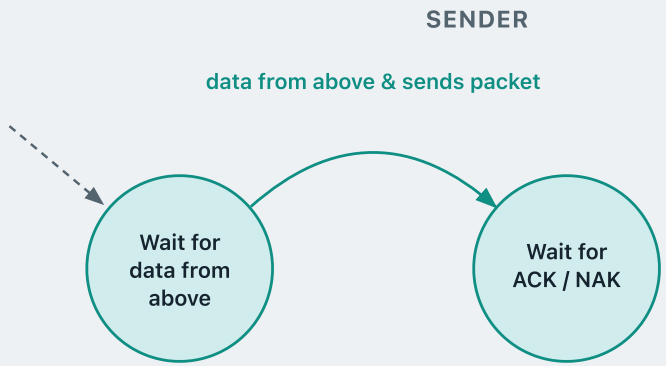
The sender's state machine has two states.

Wait for data from above — the sender is idle, waiting for the next packet from the application.

Wait for ACK / NAK — the sender has just sent a packet to the receiver, and has not heard back yet.

RDT 2.0

The protocol as a state machine

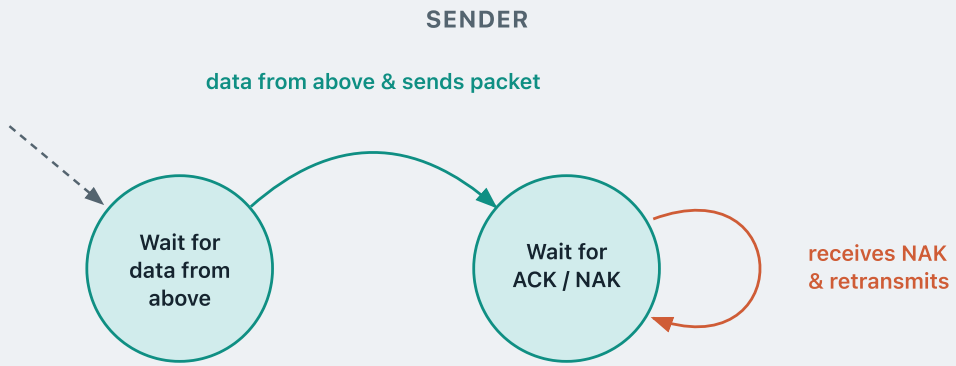


data from above & sends packet

The application hands data down. The sender builds the packet, transmits it, and moves to the waiting state.

RDT 2.0

The protocol as a state machine

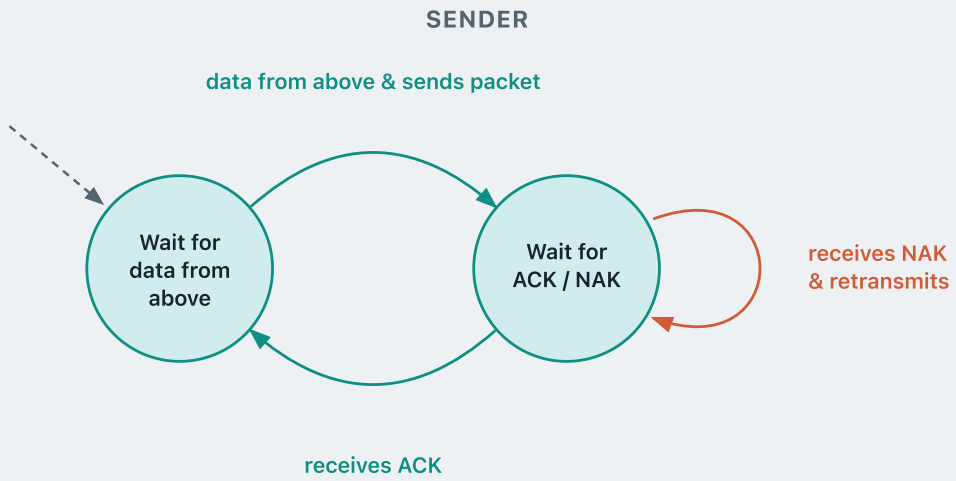


receives NAK & retransmits

The receiver reported a corrupted packet, so the sender retransmits **the same packet** — and comes straight back to the same waiting state.

RDT 2.0

The protocol as a state machine

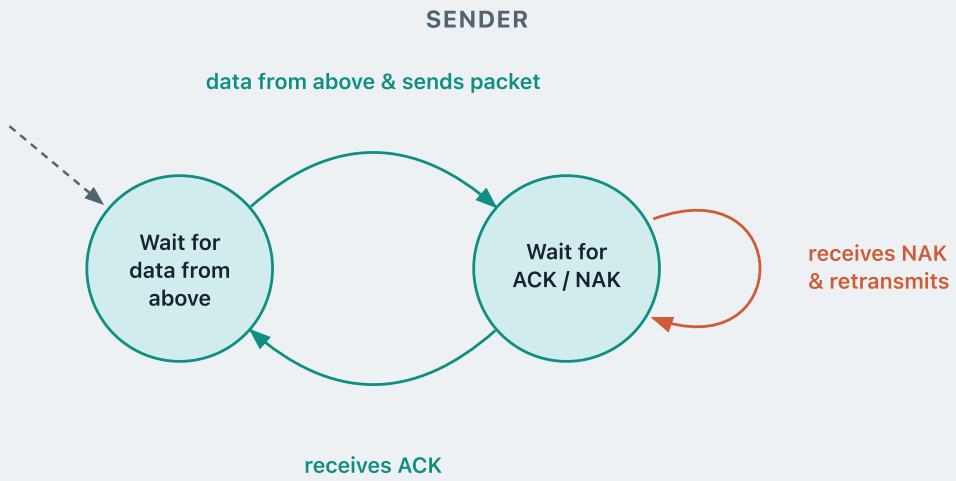


receives ACK

The packet got through. The sender returns to the idle state, ready for the next packet from above.

RDT 2.0

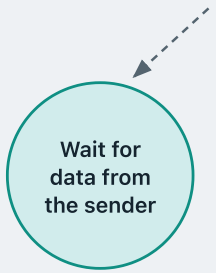
The protocol as a state machine



This is the **stop-and-wait** strategy: the sender transmits **one single packet**, then **stops** and does nothing at all until it hears back from the receiver.

RDT 2.0

The protocol as a state machine

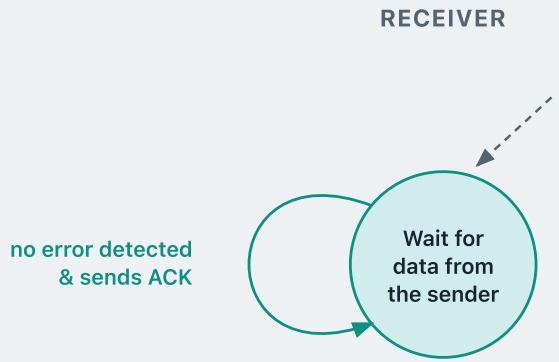


The receiver's state machine has one state.

Wait for data from the sender — it just waits for the next packet to arrive from the channel.

RDT 2.0

The protocol as a state machine

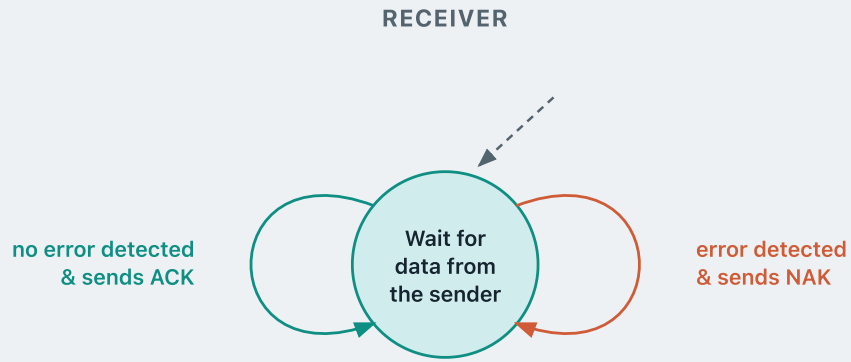


no error detected & sends ACK

Deliver the data to its application, and send an ACK back to the sender.

RDT 2.0

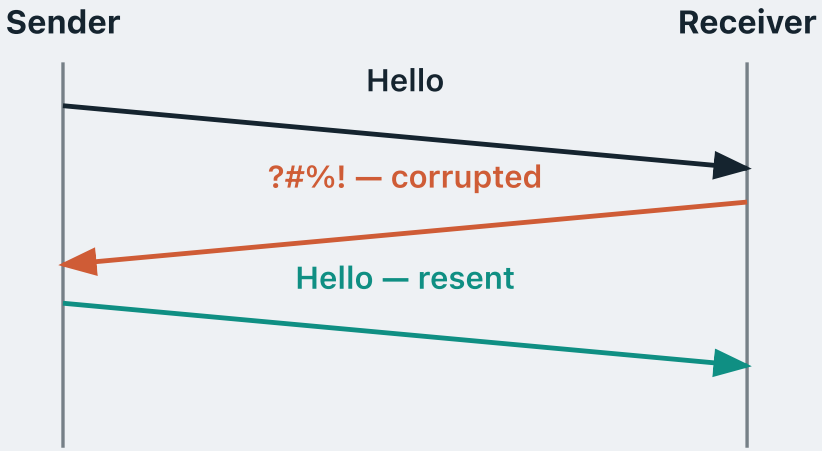
The protocol as a state machine



error detected & sends NAK

Discard the packet, and send a NAK.

A fatal flaw



We assume that the **data** sent from the sender can be corrupted. But the **ACK and NAK** messages sent by the receiver also travel back through the **same channel** — so they can also be corrupted.

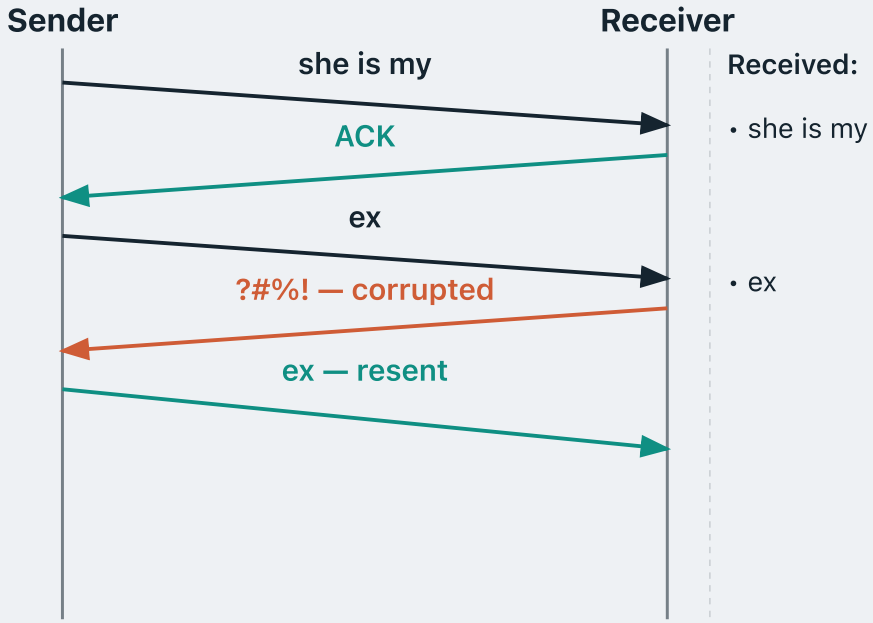
Suppose the sender sends “Hello”. The receiver gets it intact and sends back an **ACK**.

But the ACK gets **corrupted** in transit.

What should the sender do?

The sender cannot tell whether the receiver said ACK or NAK. Its only safe option is to **resend**.

A fatal flaw

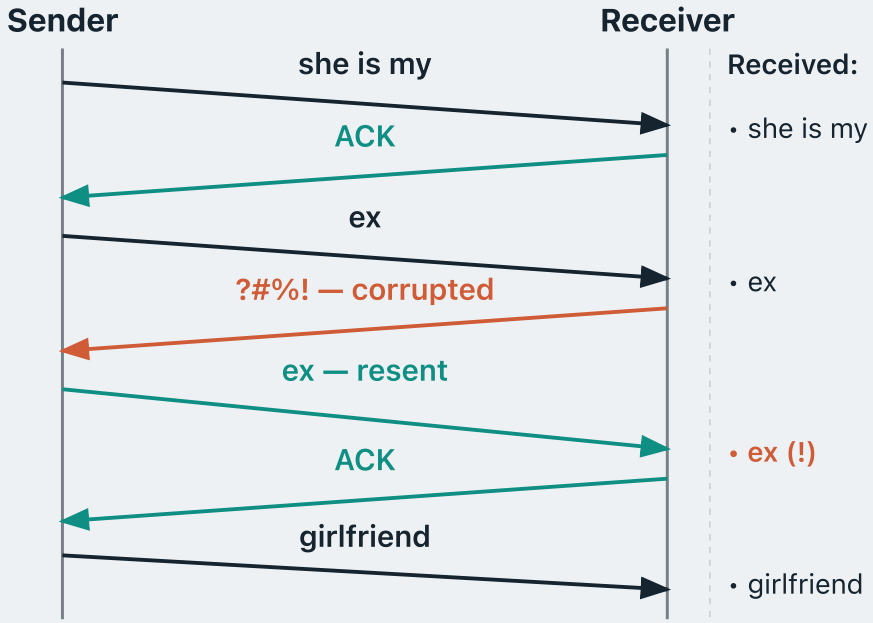


Suppose the sender wants to deliver the message **“she is my ex girlfriend”** from the application above. **“she is my”** arrives intact — receiver delivers it and sends ACK.

“ex” arrives intact — receiver delivers it and sends ACK.

The ACK gets **corrupted** — sender cannot tell, so it **resends** “ex”.

A fatal flaw



Receiver has no idea this is a resend — it **delivers "ex" again**.

"girlfriend" eventually gets through and is delivered.

In the end, the receiver delivers "she is my **ex ex** girlfriend".

This is the wrong message — it contains a **duplicate "ex"**.

How do we fix this?

RDT 2.1

The fix: a 1-bit sequence number

The fix is as simple as tagging every packet with a **sequence number**, so the receiver can tell **new data** from a **resend**.

For stop-and-wait, a **single bit** is enough — the sender alternates **0** and **1**.

SENDER

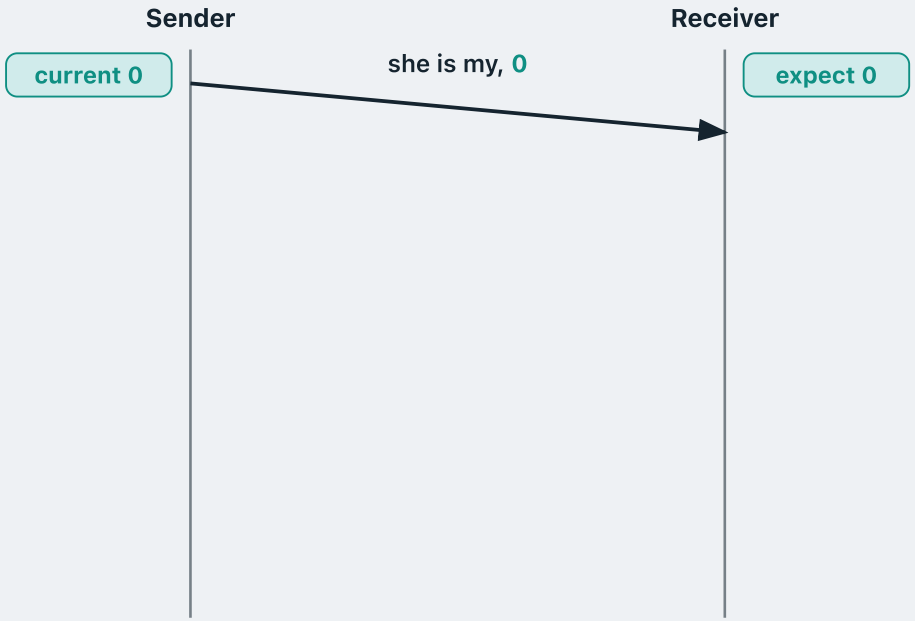
- **New data:** flip the sequence bit and transmit.
- **Retransmission:** resend with the **same** sequence number.

RECEIVER

- Packet with the **same** number as last time → **duplicate**; discard it, but re-send the ACK.
- Packet with a **different** number → **new data**; deliver it and ACK.

This protocol is known as **rdt 2.1**.

Catching a real duplicate

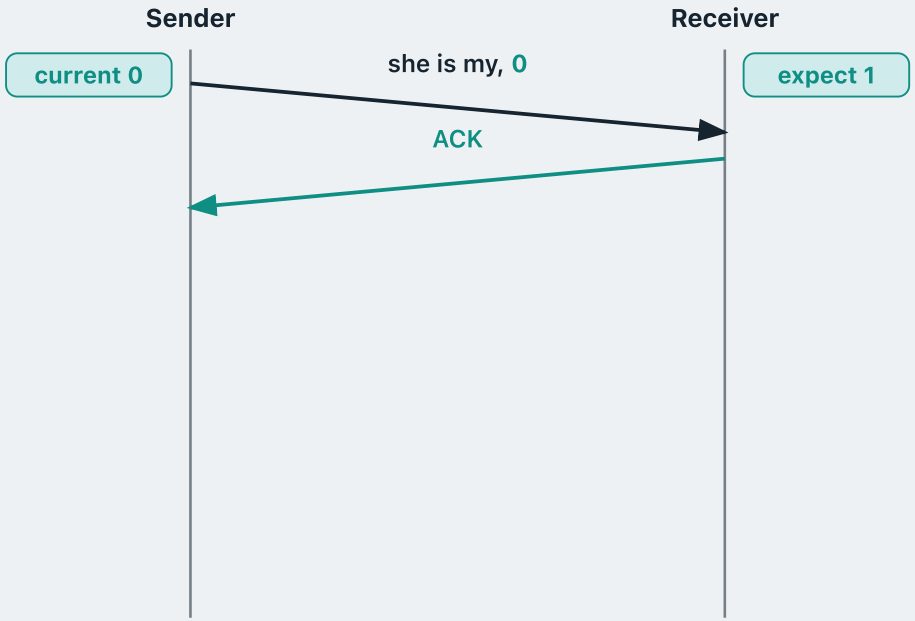


Same story as before: the sender wants to deliver **“she is my ex girlfriend”**.

Both sides start from **0**. The sender tags **“she is my”** with **0** and sends it.

0 is exactly the number the receiver is expecting, so this is **new data**.

Catching a real duplicate



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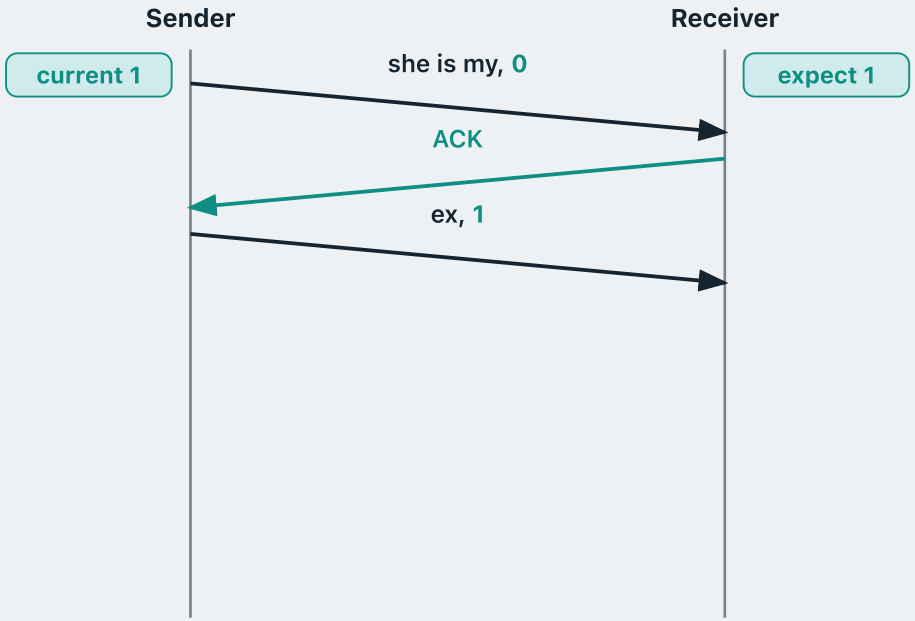
So the receiver **delivers the chunk**, sends an **ACK**, and now **expects 1**.

Catching a real duplicate



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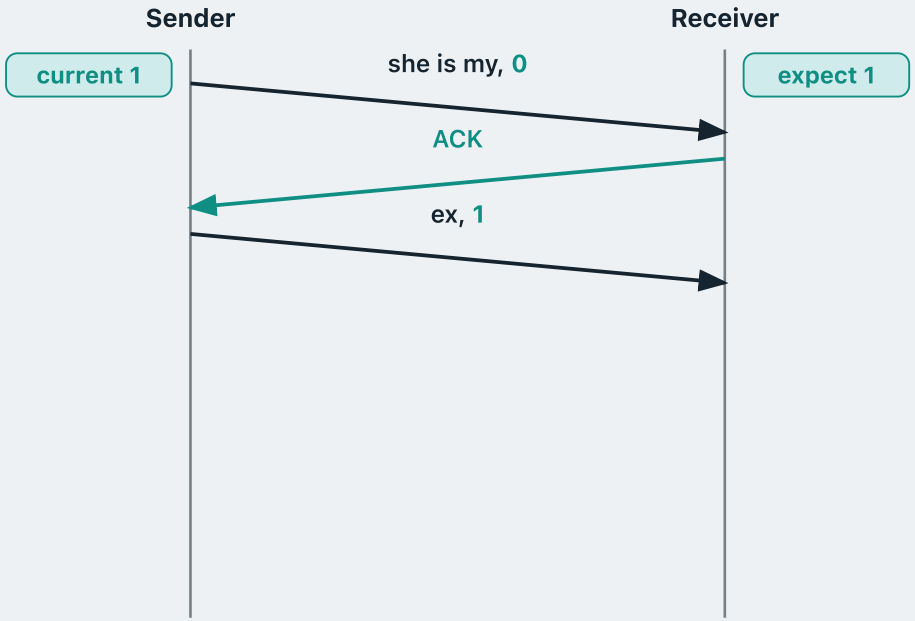
0 is exactly the number the receiver is expecting, so this is **new data**.

So the receiver **delivers the chunk**, sends an **ACK**, and now **expects 1**.

“ex” is new data too, so the sender **flips its bit** to **1** and sends it.

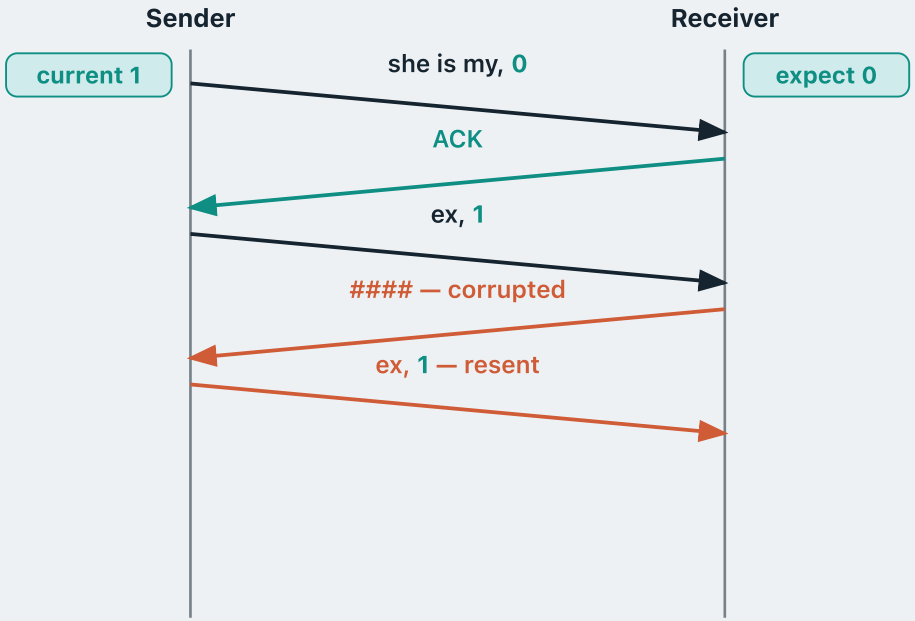
Catching a real duplicate



The receiver is **expecting 1** and this packet carries **1** — so this is **new data**.

RDT 2.1

Catching a real duplicate



The receiver is **expecting 1** and this packet carries **1** — so this is **new data**.

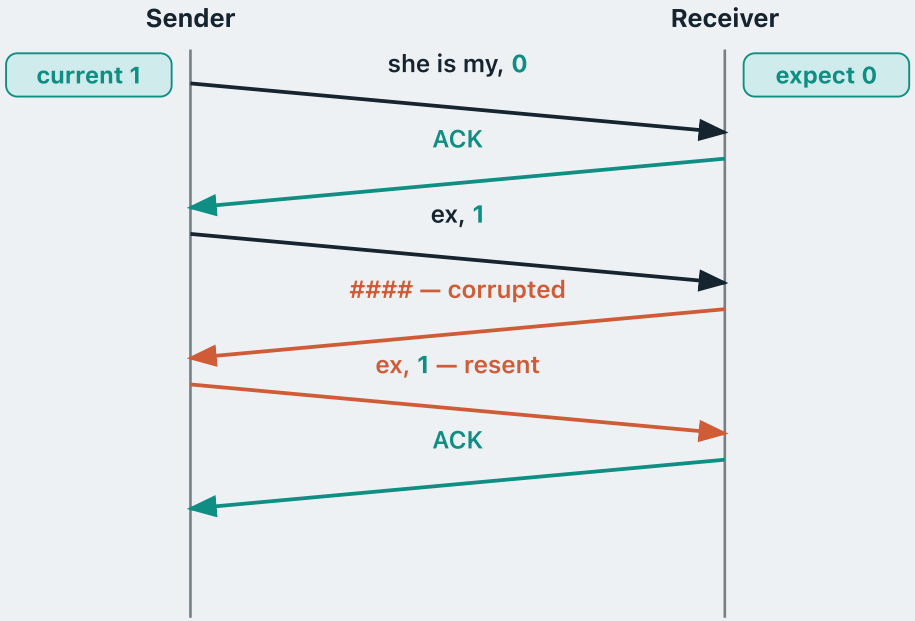
So the receiver **delivers the chunk**, sends an **ACK**, and now **expects 0**.

But this time that **ACK comes back corrupted**.

The sender cannot tell what the receiver said, so it **must resend** — this is a **retransmission**.

A retransmission is not new data, so the sender's current bit **stays at 1**: it sends **"ex, 1"** again.

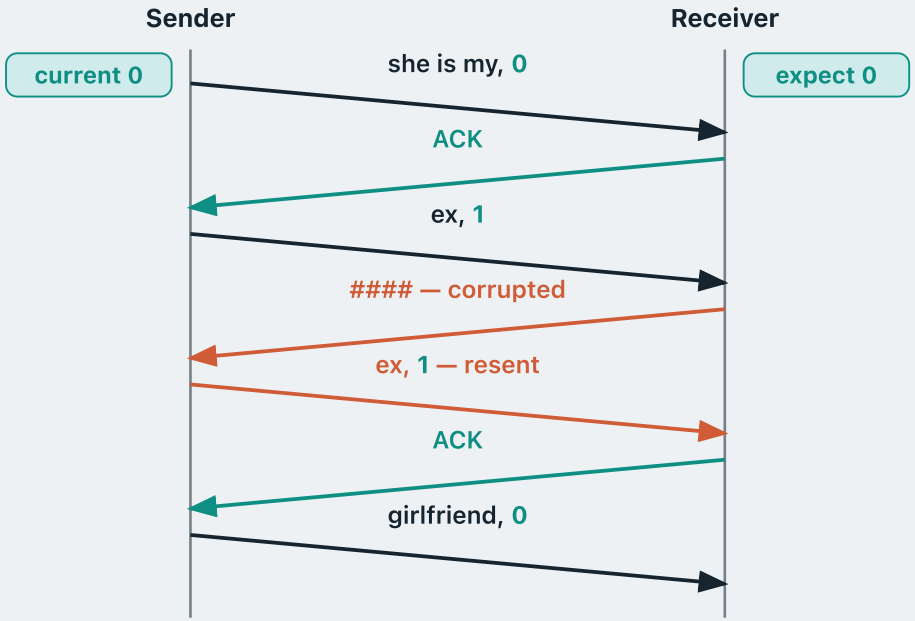
Catching a real duplicate



The receiver is **expecting 0**, but this packet carries **1** — so it is a **copy it already has**.

It **discards the copy** and just re-acknowledges, keeping its expected number at **0**.

Catching a real duplicate



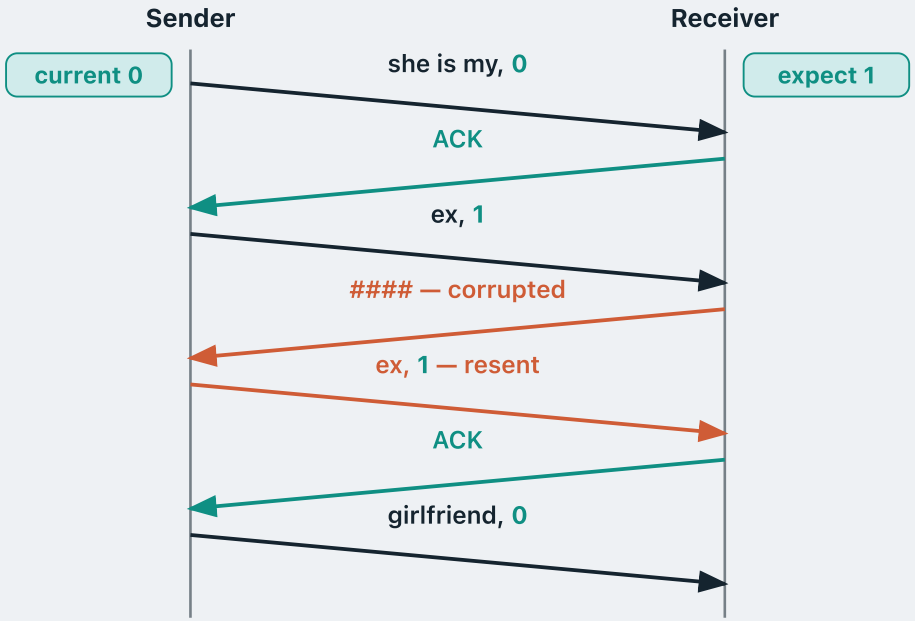
The receiver is **expecting 0**, but this packet carries **1** — so it is a **copy it already has**.

It **discards the copy** and just re-acknowledges, keeping its expected number at **0**.

The sender sees that ACK, so it can advance: it **flips to 0** and sends **"girlfriend"** with **0**.

0 is what the receiver expects, so this is **new data**.

Catching a real duplicate



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It **discards the copy** and just re-acknowledges, keeping its expected number at **0**.

The sender sees that ACK, so it can advance: it **flips to 0** and sends **"girlfriend"** with **0**.

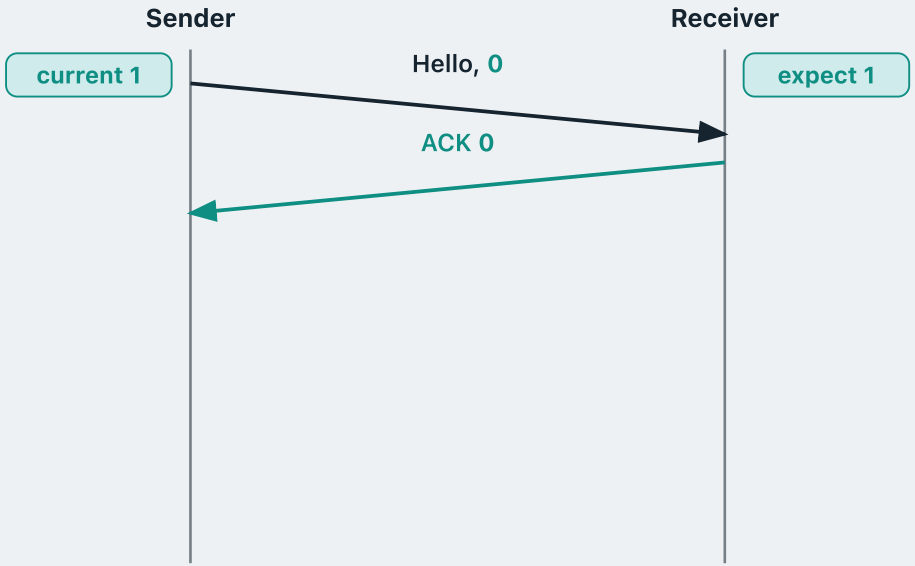
0 is what the receiver expects, so this is **new data**.

The receiver **delivers the chunk** and now **expects 1**.

"she is my ex girlfriend" ✓

Delivered correctly, with **no duplicate**.

Numbering the ACKs too

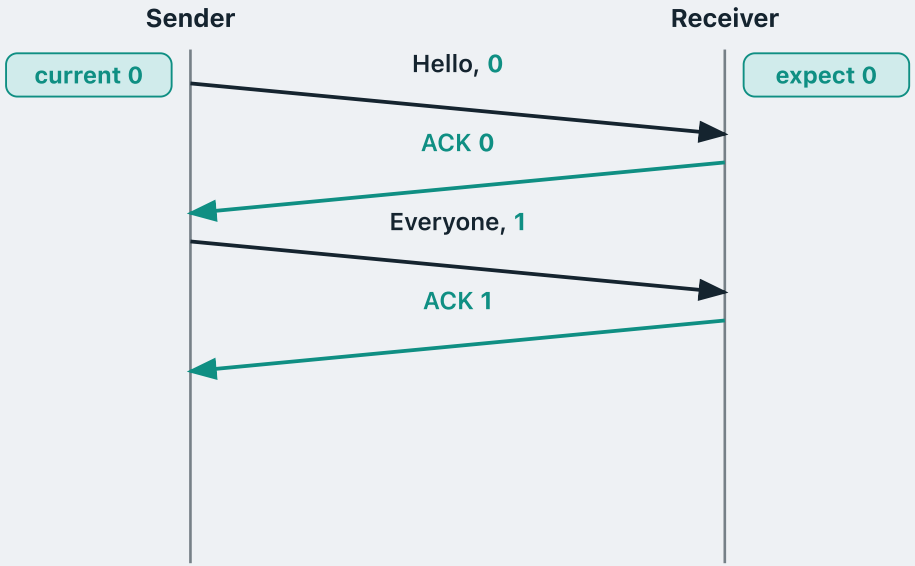


rdt 2.2 adds a neat trick: number the **receiver's** messages too.

Every ACK now carries a sequence number. **ACK 0** means "I received your packet **0**".

RDT 2.2

Numbering the ACKs too



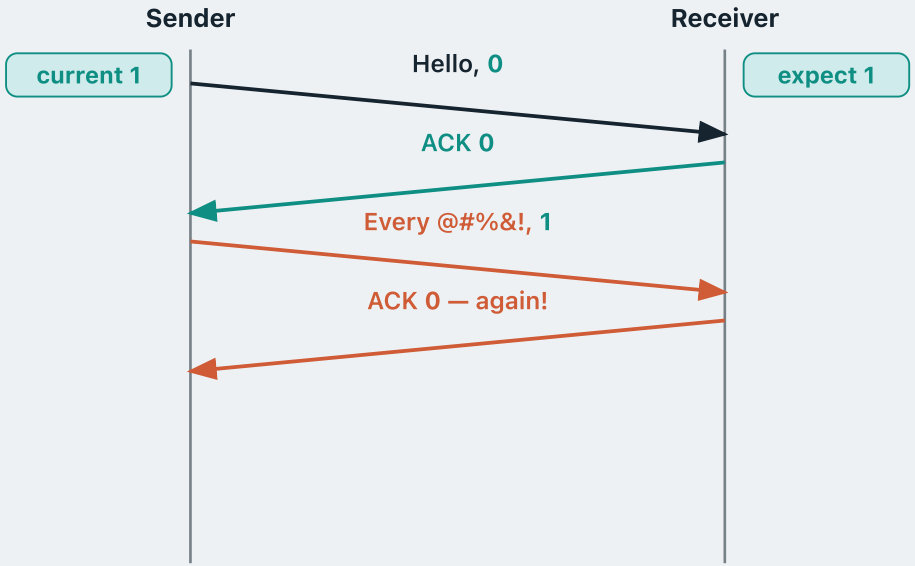
rdt 2.2 adds a neat trick: number the **receiver's** messages too.

Every ACK now carries a sequence number. **ACK 0** means "I received your packet **0**".

With nothing corrupted, the exchange simply alternates.

RDT 2.2

What if a packet is corrupted?



Now suppose that second packet arrives **corrupted**.

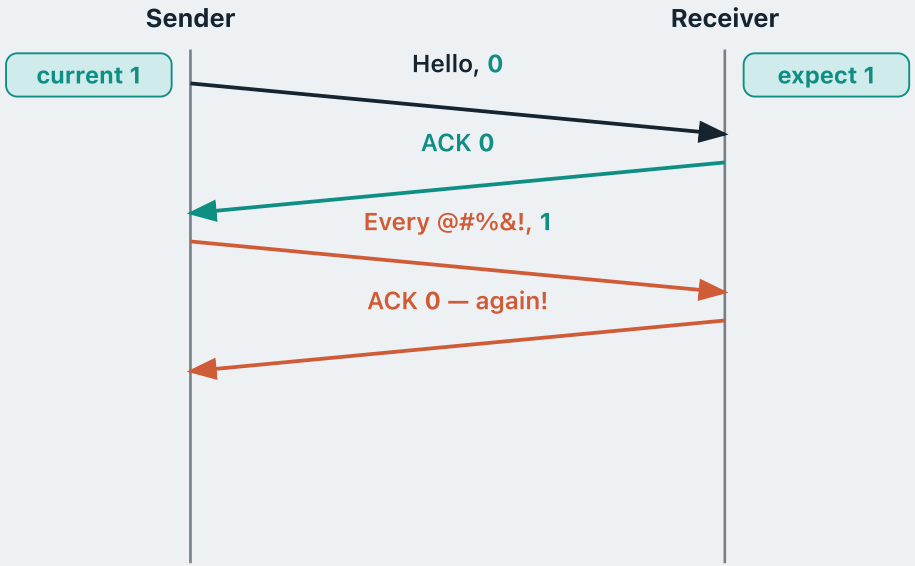
Normally the receiver would send a **NAK** — "I detected a corrupted packet, please send it again".

But its messages are numbered now. So instead of a NAK it just sends an ACK carrying the **previous** sequence number: **ACK 0**, again.

> Which says exactly this: "I still only have packet **0**. I have not received packet **1**."

RDT 2.2

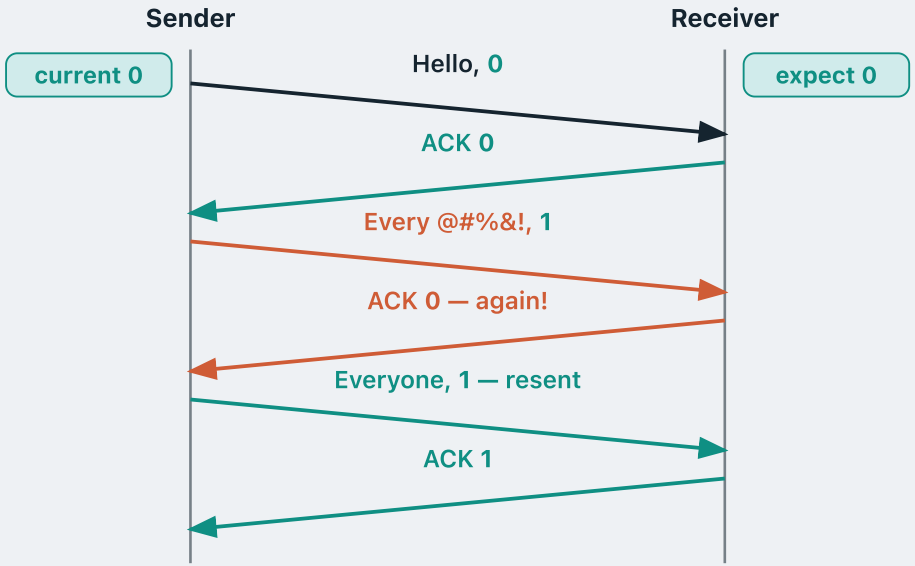
A duplicate ACK *is* a NAK



The sender is waiting for **ACK 1**. Getting **ACK 0** again tells it packet **1** never arrived — exactly what a NAK would have said.

> Seeing the same ACK number twice is called a **duplicate ACK**, and it does a NAK's job: it asks the sender to retransmit.

A duplicate ACK *is* a NAK



The sender is waiting for **ACK 1**. Getting **ACK 0** again tells it packet **1** never arrived — exactly what a NAK would have said.

> Seeing the same ACK number twice is called a **duplicate ACK**, and it does a NAK's job: it asks the sender to retransmit.

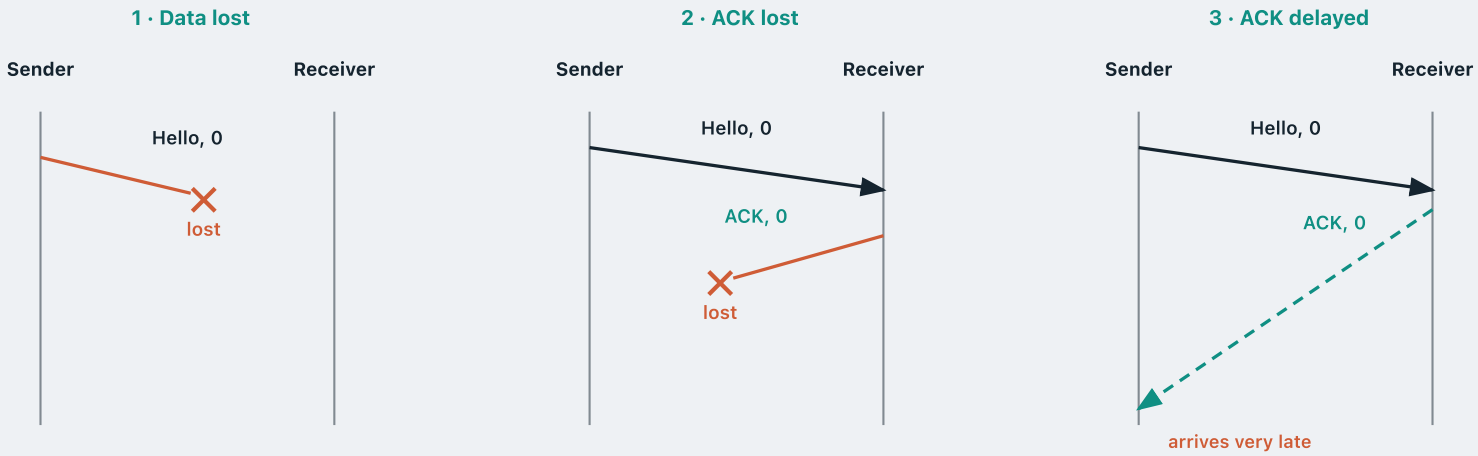
So the sender **retransmits**, using the **same sequence number** as before — **1** here.

Three new ways to fail

In **rdt 3.0** we assume the channel is even more unstable.

It might not only corrupt bits, but can also **drop** a packet completely, or **delay** any packet significantly.

More specifically, three things can happen:

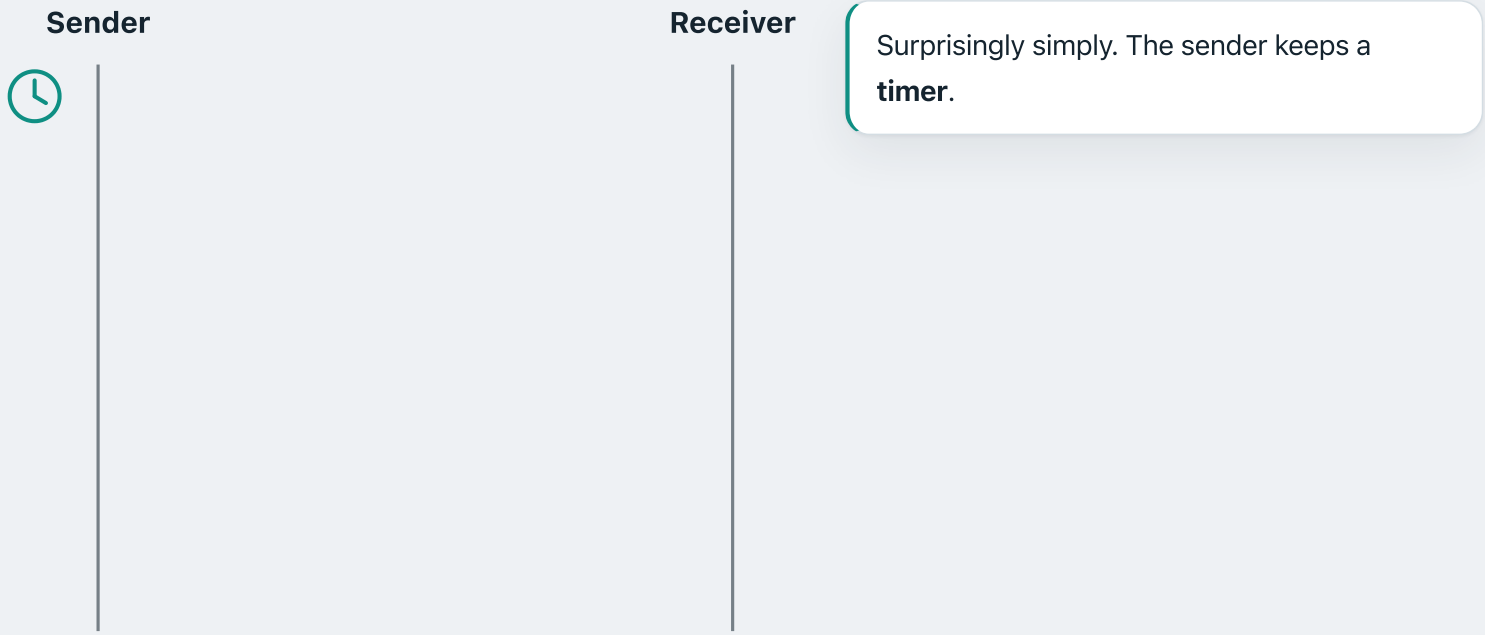


RDT 3.0

Timeout and retransmission

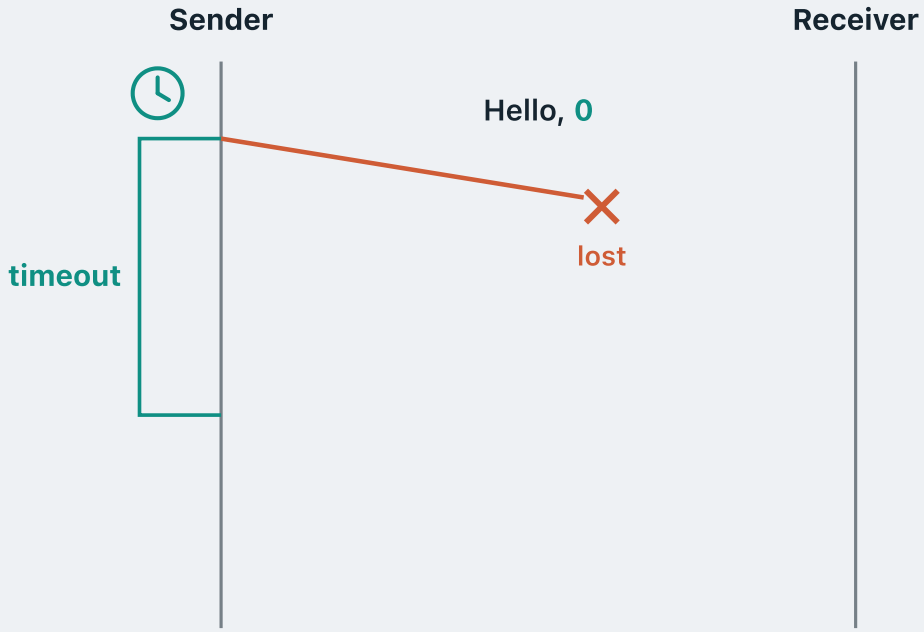
How do we resolve these problems?

Timeout and retransmission



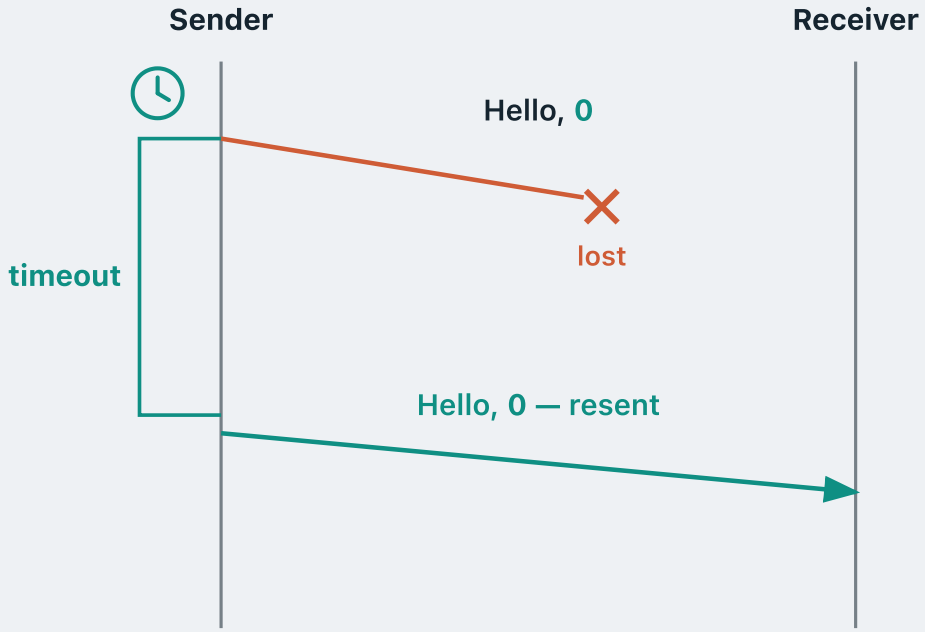
Surprisingly simply. The sender keeps a **timer**.

Timeout and retransmission



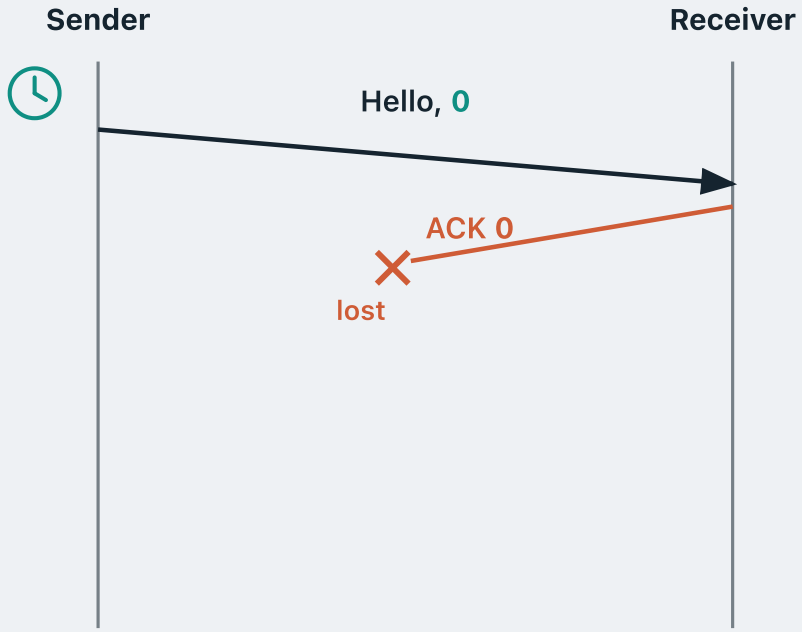
After a while the timer **expires**.

Timeout and retransmission



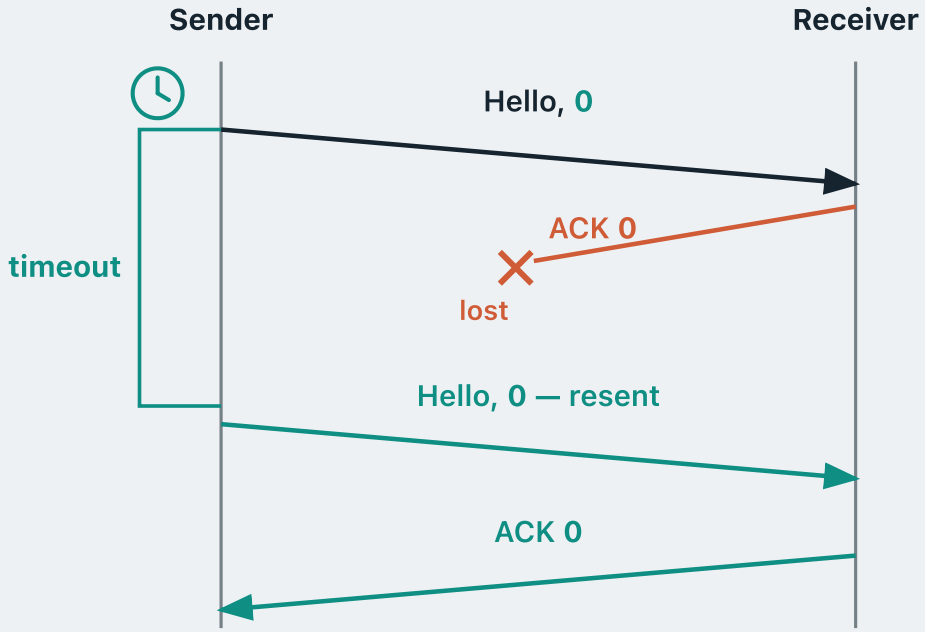
The sender assumes the worst — the packet is gone — and **retransmits it**, with the **same sequence number**.

Timeout and retransmission



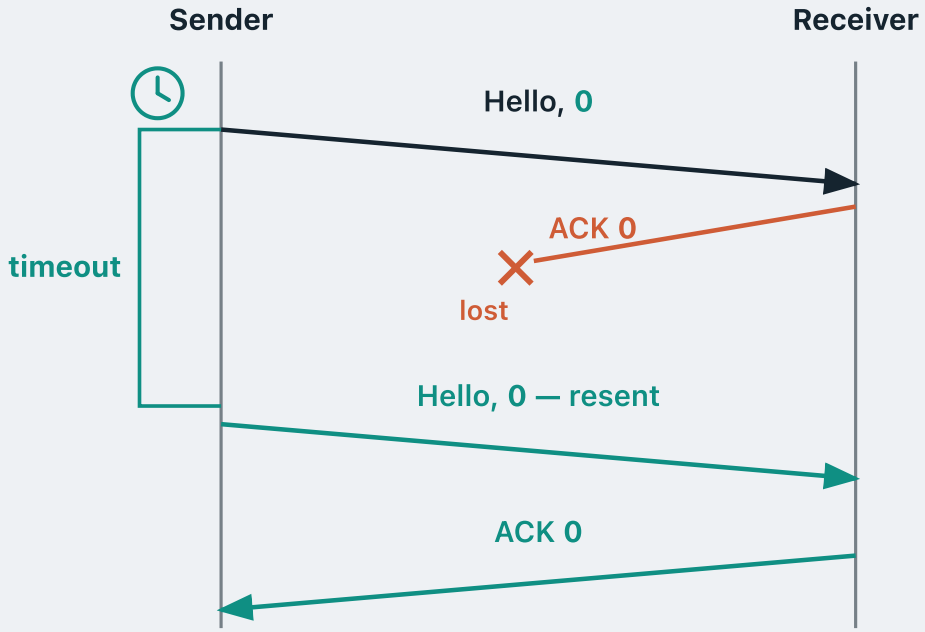
Now a different case: the data **does** arrive, but the **ACK is lost** on the way back.

Timeout and retransmission



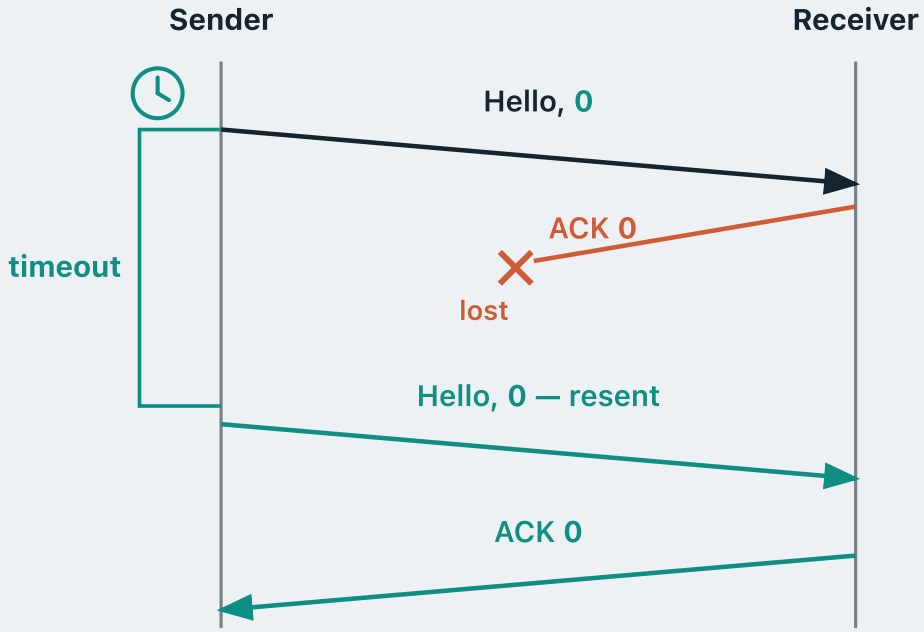
After the timer expires, the sender **still assumes the worst** — that the data was lost — and resends the **same packet** again.

Timeout and retransmission



The same happens if the ACK is merely **delayed** rather than lost: as long as the timer expires with no ACK heard, the sender always assumes the worst and **retransmits**.

Timeout and retransmission



But **how long should the sender wait?**

Too short

The ACK was merely slow, so the sender resends for nothing — **duplicate packets**.

Too long

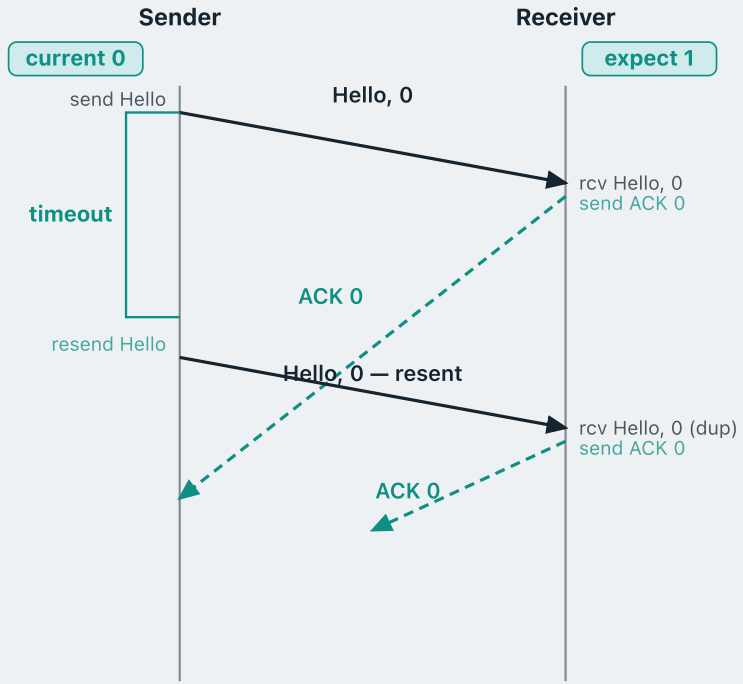
The sender sits idle long after the packet was lost — **wasted throughput**.

About **one RTT** — the time for a packet to reach the receiver and its ACK to come back.

We will come back to how that interval is chosen when we look at TCP.

A worked example

The sender wants to deliver two messages to the receiver: **Hello**, then **Everyone**.



Suppose Hello, 0 has already been sent. ACK 0 is on its way — but it is delayed.

Timer fires before ACK 0 arrives. What does the sender do?

> The **rdt 3.0 sender rule**: when the timer expires with no ACK heard, the sender assumes the worst and **retransmits the same packet**.
>
> So Hello, 0 goes out again, still numbered **0**.

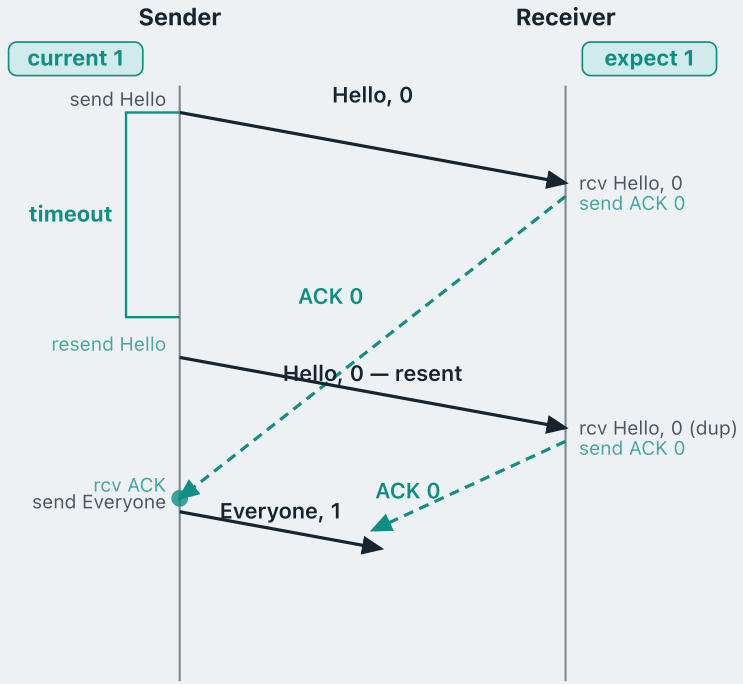
The receiver gets Hello, 0 a second time. What does it do?

> **rdt 2.2 receiver rule**: it expects **1**, this carries **0** — a duplicate. Discard it, and **ACK 0** goes back again.

The first ACK 0 arrives. What does the sender do?

A worked example

The sender wants to deliver two messages to the receiver: **Hello**, then **Everyone**.



Suppose Hello, 0 has already been sent. ACK 0 is on its way — but it is delayed.

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> The **rdt 3.0 sender rule**: when the timer expires with no ACK heard, the sender assumes the worst and **retransmits the same packet**.
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> So Hello, 0 goes out again, still numbered **0**.

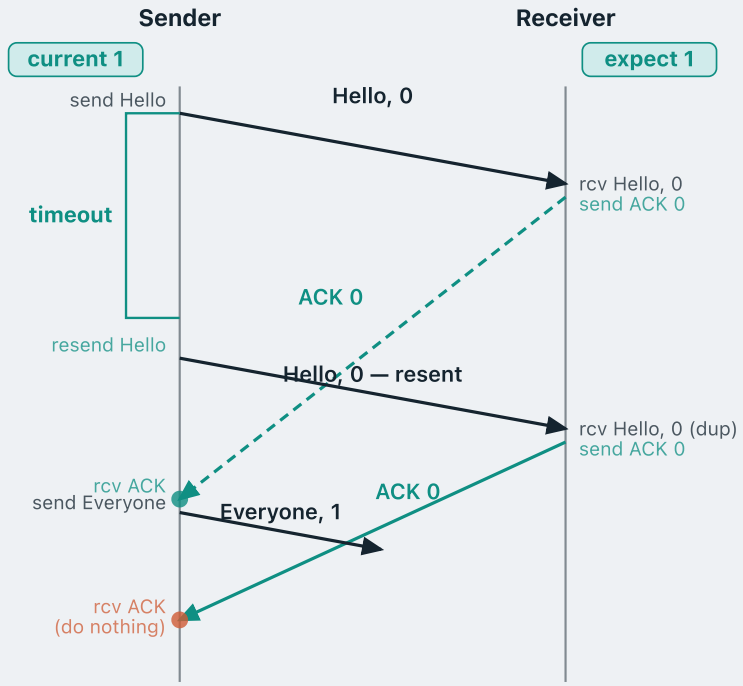
The receiver gets Hello, 0 a second time. What does it do?

> **rdt 2.2 receiver rule**: it expects **1**, this carries **0** — a duplicate. Discard it, and **ACK 0** goes back again.

The first ACK 0 arrives. What does the sender do?

> Move on — send the next packet, Everyone, 1 (flipped sequence number), and restart the timer.

What should a duplicate ACK do here?



The second ACK 0 now arrives. What should the sender do?

By the **rdt 2.2** rule, a second ACK 0 is a **duplicate ACK** — a NAK in disguise. So the sender should **retransmit Everyone, 1**.

But here that would be **wrong**.

This ACK 0 acknowledges the **previous** message, Hello, 0. It is a duplicate only because the first ACK 0 was **delayed**.

It was never meant to say that Everyone, 1 was corrupted, so it must **not** be treated as a NAK.

THE RETRANSMISSION RULE IN RDT 3.0

The sender retransmits **only when its timer expires**. Any ACK that is corrupt, or carries the wrong sequence number, is **ignored**.

That is the only difference from rdt 2.2 — the receiver is unchanged.

RECAP · STOP-AND-WAIT

What we covered

- **rdt 1.0** — over a perfectly reliable channel. No error handling needed: just send and forget.
- **rdt 2.0** — over a channel with bit errors. The receiver sends **ACK** or **NAK**; the sender retransmits on NAK.
- **rdt 2.1** — adds a **1-bit sequence number** to distinguish a resent packet from a new one.
- **rdt 2.2** — drops NAKs altogether. The receiver ACKs the **sequence number** it accepted, so a duplicate ACK is a NAK in disguise.
- **rdt 3.0** — over a lossy channel. A **timer** triggers retransmission if no expected ACK arrives in time.

MODULE 4 · CONCEPT 3 OF 4

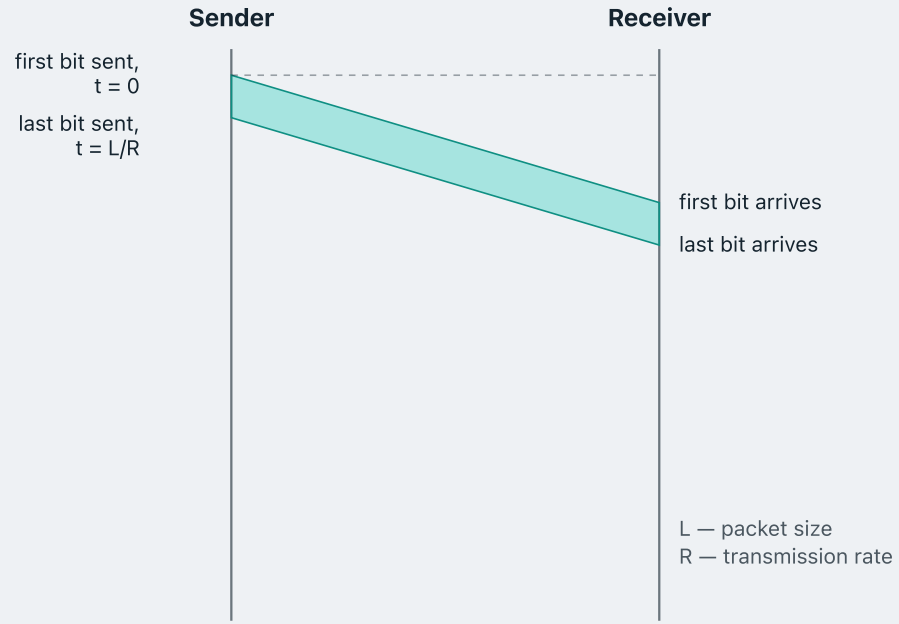
Go-back-N

Pipelined reliable data transfer

1 · Reliable data transfer | 2 · Stop-and-wait RDT | **3 · Go-back-N** | 4 · Selective repeat

PERFORMANCE

The cost of stopping and waiting



Every protocol we have built, rdt 2.0 through rdt 3.0, is **stop-and-wait**: send one packet, then sit idle until its ACK comes back.

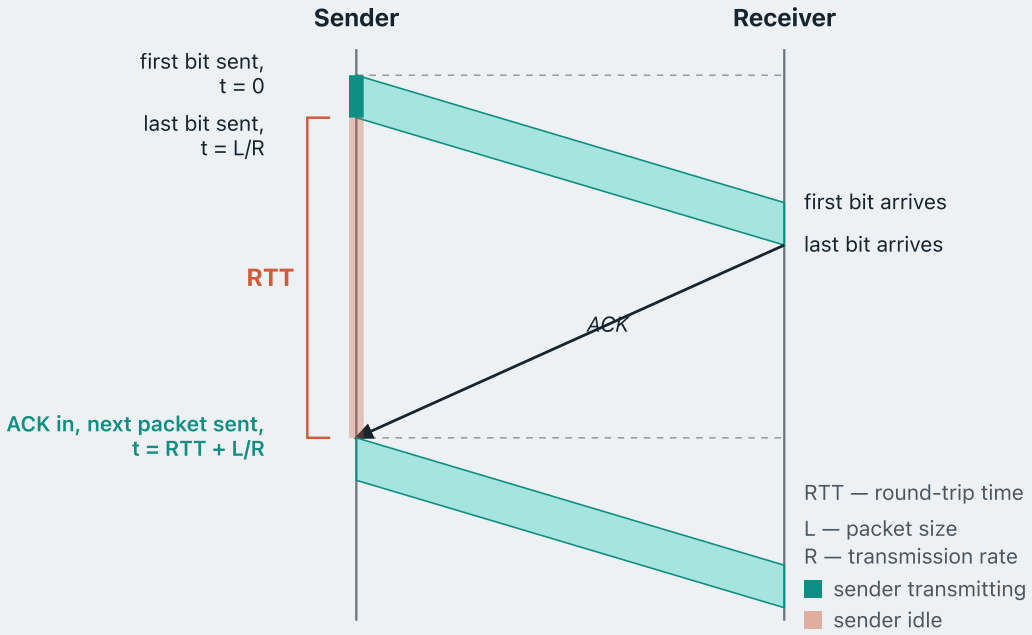
That idle time is not free. Let us try to measure what it costs us.

Say every packet is **L** bits and the sender transmits at **R** bits per second.

Putting one packet onto the channel then takes **L/R** seconds.

PERFORMANCE

How much of that time is the sender busy?



After that the sender must wait a full **RTT** for the ACK to come in.

Only then may it send again.

Therefore, sending one packet of length **L** takes the sender **RTT + L/R** seconds.

Of that, the sender is busy for only the first **L/R**, while it is transmitting. Every moment after that it is **idle**.

Fraction of the time the sender is busy:

$$U = (L/R) \div (RTT + L/R)$$

PERFORMANCE

How bad is it? A worked example

Take a 1 Gbps link, an RTT of 30 ms, and 1 KB packets.

$$L/R = 8000 \text{ bits} \div 10^9 \text{ bits/s} = 8 \text{ µs}$$

$$U = 0.008 \div (30 + 0.008) = 0.00027$$

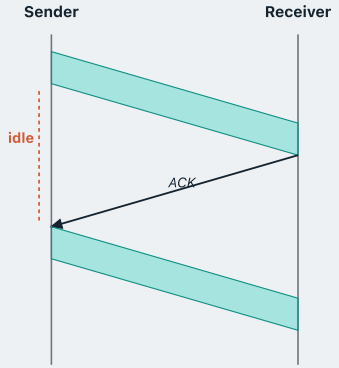
> The sender is busy **0.027%** of the time. Put another way: a **gigabit** link delivers about **267 kbps** — roughly a thousandth of its capacity.

So how do we fix this?

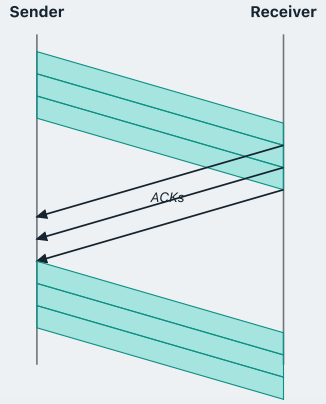
PIPELINING

Send several before you wait

Stop-and-wait — one at a time



Pipelined — several in flight



Recall the stop-and-wait utilisation: $U_{SW} = (L/R) \div (RTT + L/R)$

The channel has not changed, so the first ACK still returns at **RTT + L/R** — same denominator. But in that one window the sender has now sent **three** packets, so it was busy for **3(L/R)**:

$$U_{pipe} = 3(L/R) \div (RTT + L/R)$$

Three times the work in the same round trip — **3× more efficient**, as long as the three packets still fit inside it.

PIPELINING

What pipelining demands in return

Pipelining is not free. Two things have to grow.

A bigger sequence space

In stop-and-wait only one packet is ever in flight, so a single bit is enough: the receiver only has to tell “this one” from “the next one”.

With pipelining, however, we may have several packets in flight at once. If we pipeline three, the receiver has to tell the **first** from the **third** — one bit cannot do that.

Buffering

In a pipeline we have **multiple packets in flight** — already sent, but not yet acknowledged — and every one of them is liable to be retransmitted.

The sender therefore cannot just send and forget: it must remember every packet in flight, which is why it has to buffer **all** of them.

We will now learn about two pipelined protocols: **go-back-N** and **selective repeat**.

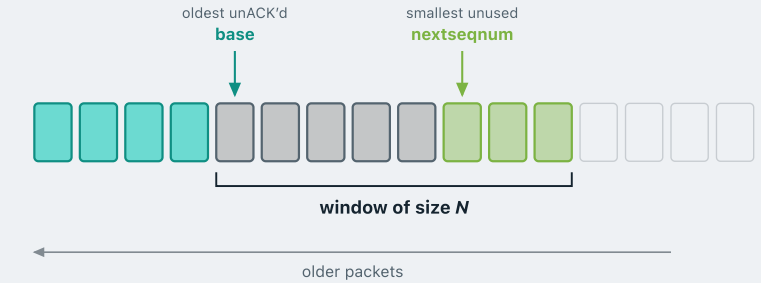
GO-BACK-N

Go-back-N

- The sender keeps a **window** of up to *N* unacknowledged packets in flight at once.
- If any packet is lost or corrupted, the sender resets its progress — it **goes back** to that packet and retransmits it along with everything that follows — hence the name.
- What does the receiver do with the packets that arrived after the lost one? Simple: it **discards them**. It does not buffer any out-of-order packets.

GO-BACK-N

The sliding window



- acknowledged
- sent, not yet acknowledged
- usable, not yet sent
- not usable yet

The data the sender works with: a stream of packets fed from the application layer, each rectangle one packet.

Acknowledged — packets the receiver has confirmed. The sender has delivered these and no longer needs to keep them.

Window of size *N* — the sender keeps at most *N* unACK'd packets in flight at once.

Sent, not yet ACK'd — packets currently in transit. The sender holds these in its buffer in case a retransmission is needed.

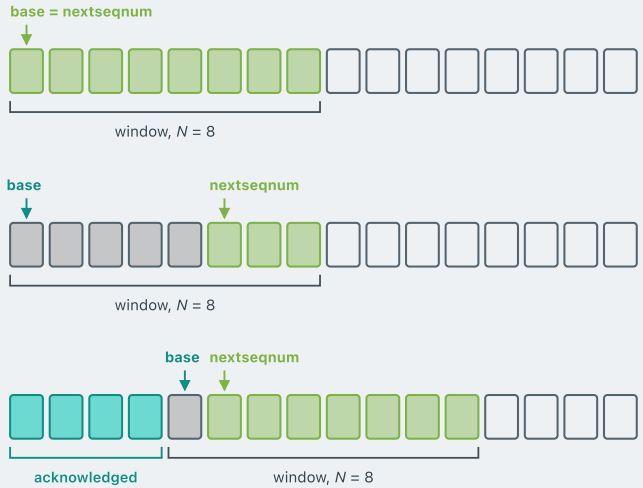
Usable, not yet sent — within the window, so the sender can transmit these now.

base — points to the oldest unACK'd packet. The window begins here.

nextseqnum — the next packet to transmit.

GO-BACK-N

The window in motion



Nothing sent yet. The whole window is usable, and `base` and `nextseqnum` sit together.

Called from above. Each time the layer above (application layer) hands data to the sender, the sender transmits it and advances `nextseqnum`. When `nextseqnum = base + N`, the window is full and no more data can be sent.

An ACK arrives. `base` advances past everything acknowledged and the window slides right, opening up sequence numbers that were not usable before.

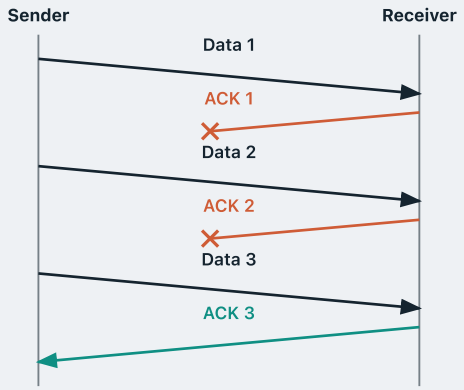
GO-BACK-N

Cumulative acknowledgements

An **ACK *n*** means: **every** packet up to and including *n* has been received correctly — not just packet *n*, but everything before it too. That is what makes it **cumulative**.

Suppose **ACK 1** and **ACK 2** are both lost in transit.

ACK 3 arrives — and it alone tells the sender that 1, 2 and 3 all got through, even though ACK 1 and ACK 2 were lost.



GO-BACK-N

The receiver's rule in Go-back-N, informally

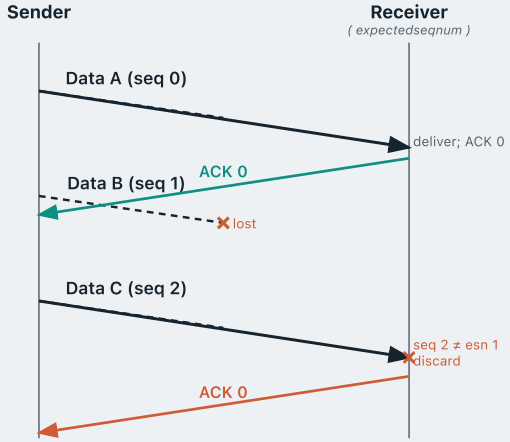
- The receiver keeps a **state**: a single number, `expectedseqnum` — the sequence number of the **next in-order** packet it expects. (e.g. if it has delivered 4, 5, 6 → `expectedseqnum = 7`)

- If the arriving packet **is** the expected one: **deliver** it and send an **ACK**.

- Otherwise: **discard** the packet and **re-ACK** the last in-order sequence number. The receiver keeps **no buffer**.

GO-BACK-N

The receiver in action



The sender pipelines three packets simultaneously — **Data A** (seq 0), **Data B** (seq 1), **Data C** (seq 2) — all in flight at once.

The receiver is waiting for the first in-order packet.

`expectedseqnum = 0`

Data A (seq 0) arrives. Seq 0 = expectedseqnum — delivered and acknowledged.

`expectedseqnum: 0 → 1`

Data B (seq 1) is **lost**. The receiver never sees it.

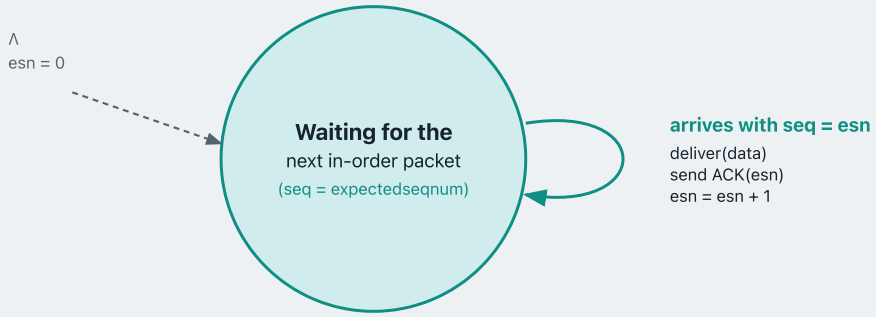
`expectedseqnum = 1`

Data C (seq 2) arrives — but seq 2 ≠ expectedseqnum 1. Out of order: **discarded**. ACK 0 re-sent.

`expectedseqnum = 1`

GO-BACK-N

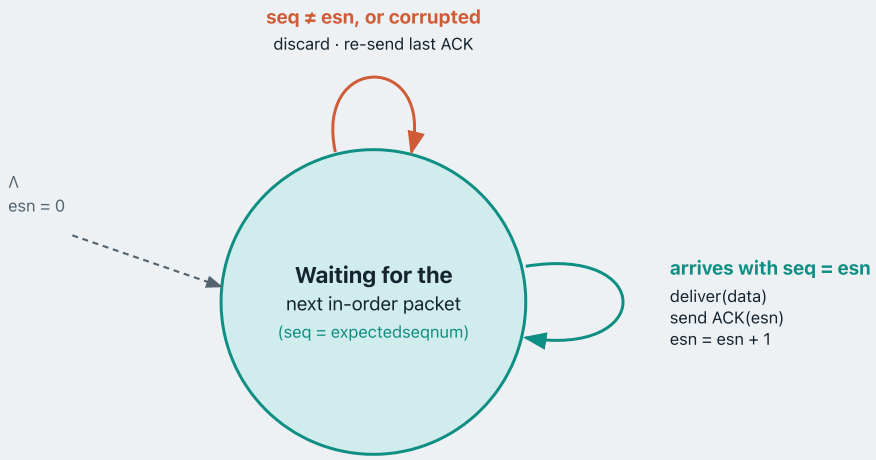
Receiver: the state machine



In-order packet:

```
if pkt.seq == expected_seq_num:
    deliver(pkt.data)
    send_ack(expected_seq_num)
    expected_seq_num = (expected_seq_num + 1) % seq_space
```

Receiver: the state machine

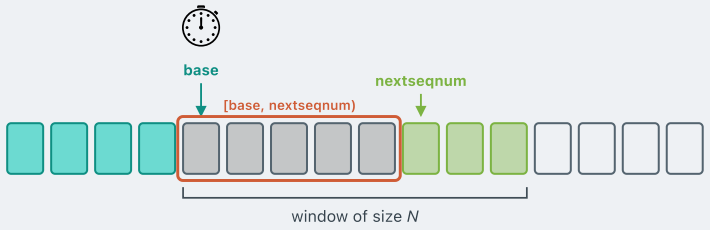


Out of order or corrupted:

```
else:
    discard(pkt)
    send_ack((expected_seq_num – 1) % seq_space)
```

GO-BACK-N

Timeout: go back *N*



Just like in rdt 3.0, the channel can **lose** packets — Go-back-N still needs a **timer**.

In Go-back-N, the sender uses just **one timer** for the whole window — set at the oldest unacknowledged packet, `base`.

When it expires: resend **every** packet in `[base, nextseqnum)` — **not** just the one that timed out.

The reason: if `base` itself was lost, the receiver — following Go-back-N's rule — will discard every packet that arrived after the gap. The whole window must go again.

This is why the sender keeps a **buffer** — a stored copy of every sent-but-unACK'd packet. Without it, go-back-N is impossible.

GO-BACK-N

The sender's rule in Go-back-N, informally

- The sender keeps a **window** of at most *N* unacknowledged packets, tracked with two variables: `base` (oldest unACK'd) and `nextseqnum` (next to send).

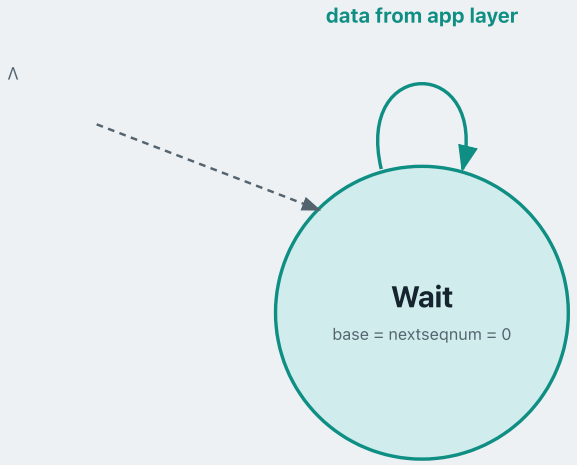
- **Call from above** — application layer requests sending a new packet.
  - If the window is not full (`nextseqnum < base + N`): assign sequence number `nextseqnum` to the packet, store it in the buffer, send it, and advance `nextseqnum`.
  - Otherwise: the window is full — refuse and wait.

- **ACK *n* arrives** — advance `base` to *n* + 1, and restart the timer (or stop it if nothing is left outstanding).

- **Timeout** — resend every buffered packet in `[base, nextseqnum)`.

GO-BACK-N

Sender: the state machine



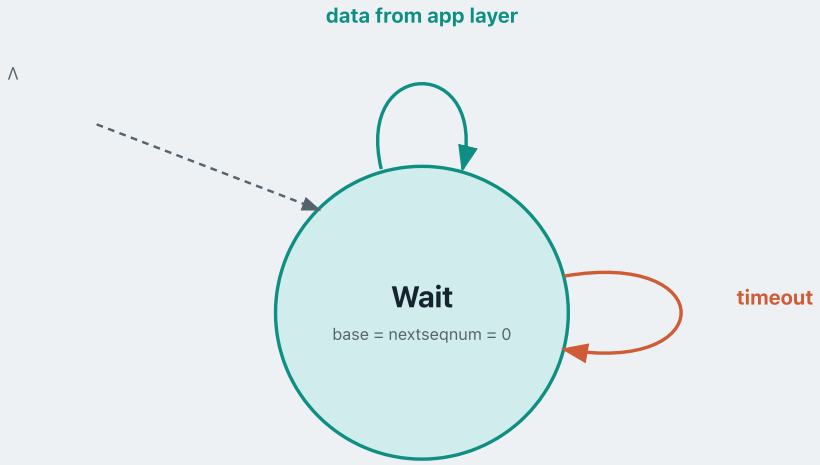
The sender sits in one **Wait** state, tracking three things: `base` (oldest unACK'd), `nextseqnum` (next sequence number to assign), and `pkt_buffer` (a copy of every sent-but-unACK'd packet).

Data from app layer:

```
if window not full  (nextseqnum < base + N):
    buffer packet at seq = nextseqnum
    send it
    if nothing was in flight → start timer
    nextseqnum += 1
else:
    reject — window full
```

GO-BACK-N

Sender: the state machine

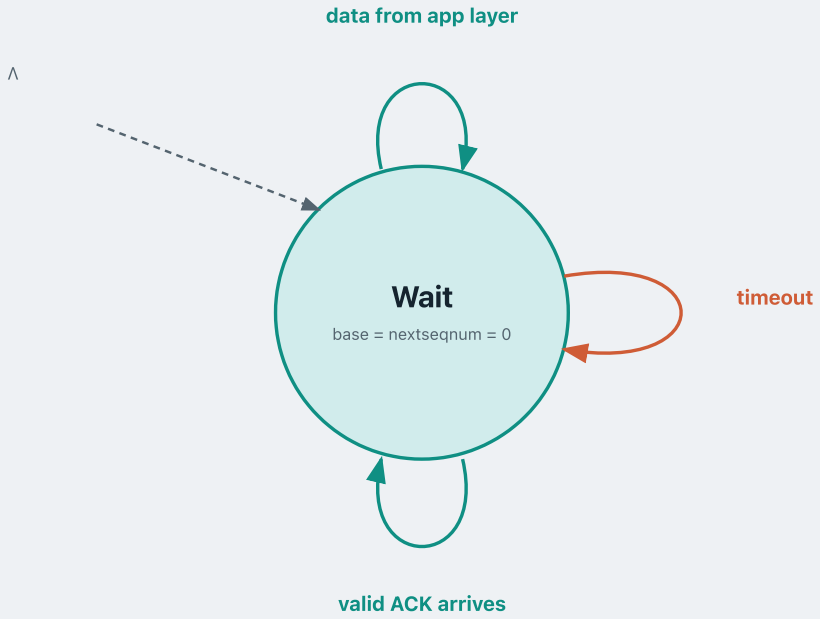


The sender sits in one **Wait** state, tracking three things: `base` (oldest unACK'd), `nextseqnum` (next sequence number to assign), and `pkt_buffer` (a copy of every sent-but-unACK'd packet).

Timer fires:

```
restart timer
for every seq in [base, nextseqnum):
    re-send buffered packet at seq
```

Sender: the state machine



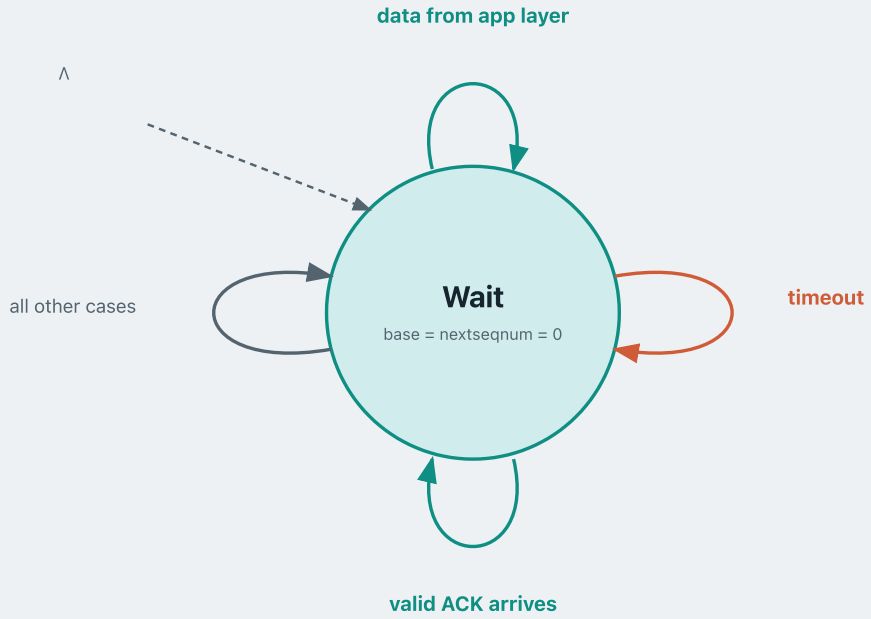
The sender sits in one **Wait** state, tracking three things: `base` (oldest unACK'd), `nextseqnum` (next sequence number to assign), and `pkt_buffer` (a copy of every sent-but-unACK'd packet).

Valid ACK — *not corrupted* AND `pkt.ack ≥ base`:

```
slide window → base = pkt.ack + 1
free buffer slots now below base
if nothing outstanding (base == nextseqnum):
    stop timer
else:
    restart timer for the new oldest
```

GO-BACK-N

Sender: the state machine



The sender sits in one **Wait** state, tracking three things: `base` (oldest unACK'd), `nextseqnum` (next sequence number to assign), and `pkt_buffer` (a copy of every sent-but-unACK'd packet).

All other cases:

- `ACK packet is corrupted`
- `ACK number < base — stale ACK`

→ `do nothing`

RECAP · GO-BACK-N

What we covered

- **Sliding window** — the sender keeps a sliding window of at most *N* unacknowledged packets in flight.
- **Receiver rule** — the receiver sends a cumulative ACK and only accepts the next expected packet (`expectedseqnum`); it does not buffer out-of-order packets.
- **Timeout** — one timer covers the entire window; when it fires, the sender retransmits every packet currently in flight.

MODULE 4 · CONCEPT 4 OF 4

Selective repeat

Pipelined reliable data transfer

1 · Reliable data transfer | 2 · Stop-and-wait RDT | 3 · Go-back-N | **4 · Selective repeat**

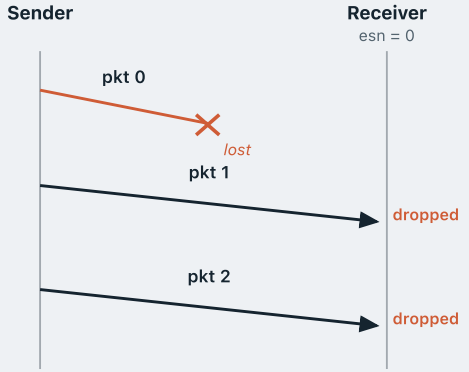
GO-BACK-N

The cost of the receiver's rule

Go-back-N wastes network capacity whenever packets arrive out of order.

Suppose pkt 0 is lost in transit. The receiver's `expectedseqnum` stays at 0.

> pkt 1 and pkt 2 arrive correctly — but because `expectedseqnum` = 0, both are **discarded**.



SELECTIVE REPEAT

Introducing Selective Repeat

The receiver buffers and ACKs every packet individually

Not cumulatively — each ACK names exactly one packet. Out-of-order packets are **buffered**, not discarded, and delivered upward once the gap is filled.

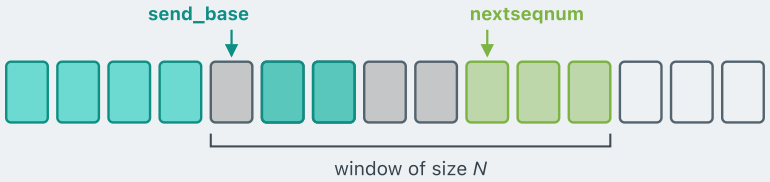
For this purpose, the receiver also keeps a **window** of packets it is waiting to accept and buffer.

The sender resends only the packets it has no ACK for

Which means it needs a **separate timer for each unacknowledged packet**.

SELECTIVE REPEAT

The sender's window in Selective Repeat



acknowledged | sent, not yet acknowledged | usable, not yet sent | not yet usable

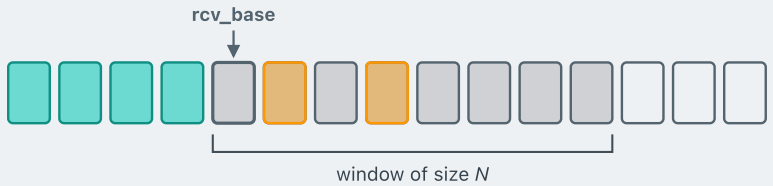
`send_base` always points to the **oldest unacknowledged** packet. The window always starts from `send_base`.

In Selective Repeat, since packets can be ACKed individually — **some slots within the window can be acknowledged while earlier ones are not** — the window can have holes.

In Go-back-N this was impossible — a cumulative ACK can never skip over an unacknowledged packet, so there could never be a hole inside the window.

SELECTIVE REPEAT

The receiver’s window in Selective Repeat



delivered to app layer not yet received received & buffered not yet acceptable

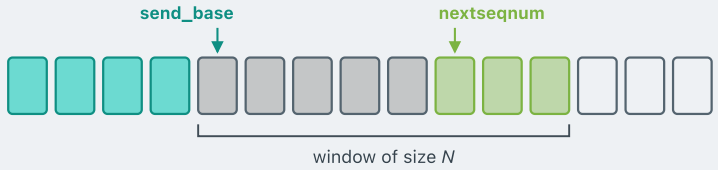
`rcv_base` points to the **first packet not yet delivered** — all packets below it have been received in order and passed to the application.

The receiver window is also of size *N* — the same as the sender’s window.

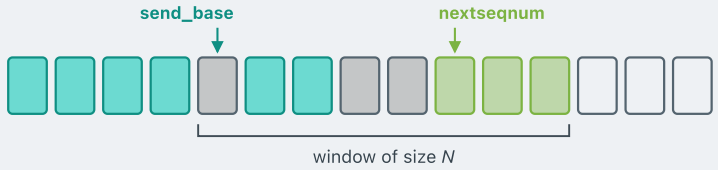
The receiver can accept and buffer any packet that falls within `[rcv_base, rcv_base+N)`, even if earlier packets have not arrived yet.

SELECTIVE REPEAT

How the sender's window slides forward

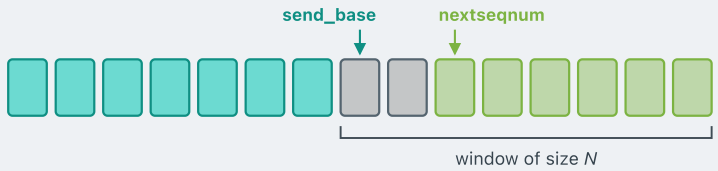


Before. Five packets are outstanding, none acknowledged yet.



ACKs arrive for two of them.

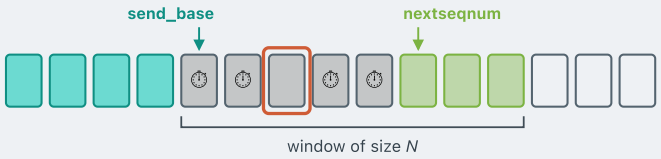
- Those slots are marked acknowledged — can we advance the window by 2?
- **No.** Since the packet at `send_base` is still outstanding, `send_base` does not move, and the window is always defined as [`send_base`, `send_base+`*N*).



Only when `send_base` itself is acknowledged does the window advance — jumping forward past every contiguous acknowledged slot at once.

SELECTIVE REPEAT

Timeout: resend just the one



Each outstanding packet has **its own timer**.

When any timer expires, the sender resends **that packet only** and restarts **that** timer. Everything else in the window is untouched.

This is in contrast with Go-back-N, where one timer covered the whole window and one expiry resent all packets in flight.

SELECTIVE REPEAT

The sender's rules

`data received from above`

If the next available sequence number falls **inside the window**, send the packet.

`timeout`

Every packet has its own timer. Resend **that one packet** and restart its timer.

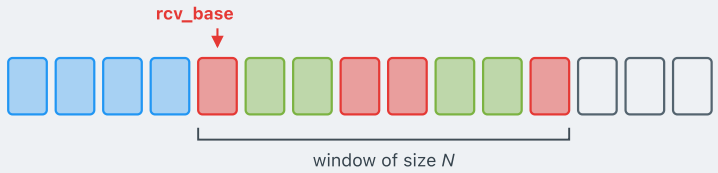
`ACK received, seq# in [send_base, send_base+N−1]`

Mark that packet as received.

If its sequence number **equals `send_base`**, advance the base to the next unacknowledged number.

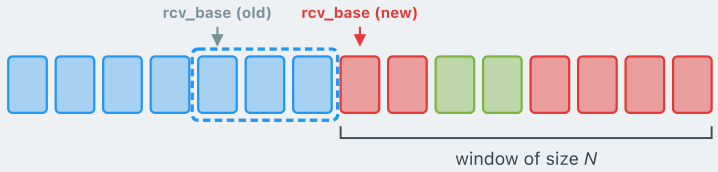
SELECTIVE REPEAT

How the receiver's window slides forward



A packet arrives within the window, but it is not at `rcv_base`.

Buffer it. Send a selective ACK for it. `rcv_base` does not move — the window stays in place, waiting for the missing packet.



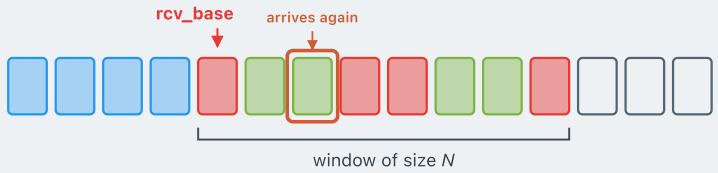
The packet at `rcv_base` arrives.

Deliver it to the application, then deliver every consecutively buffered packet above it. Advance `rcv_base` past all of them — the window jumps forward in one go.

expected, not yet received · received and buffered · delivered to application

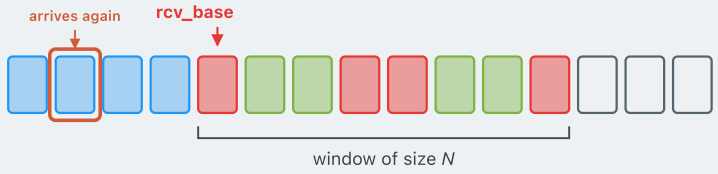
SELECTIVE REPEAT

Corner cases for the receiver



A packet that is already buffered arrives again.

- The first selective ACK clearly did not get through. The sender **times out** on this one, and retransmits.
- **Ignore** the duplicate. Re-send the **lost selective ACK**.



A packet from *below* the receiver's window arrives again.

- The selective ACK for this old packet clearly did not get through — the sender timed out and retransmitted.
- **Re-send the lost ACK**.

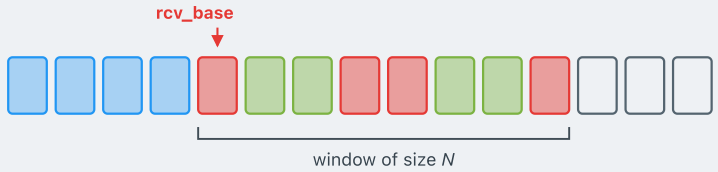
expected, not yet received received and buffered delivered to application

SELECTIVE REPEAT

The receiver's rules

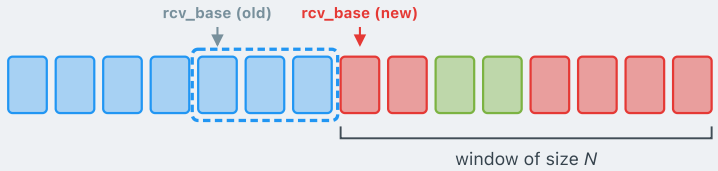
When a new packet arrives, look at its seq#, then:

expected, not yet received buffered delivered



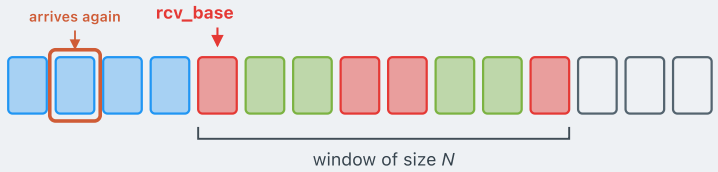
seq# in [`rcv_base`, `rcv_base+N−1`]

- Send a **selective ACK**.
- Out of order and not yet buffered? **Buffer it.**



Equals `rcv_base`?

- Deliver it, together with any consecutively buffered packets above it. **Advance the window** past all of them.



seq# in [`rcv_base−N`, `rcv_base−1`]

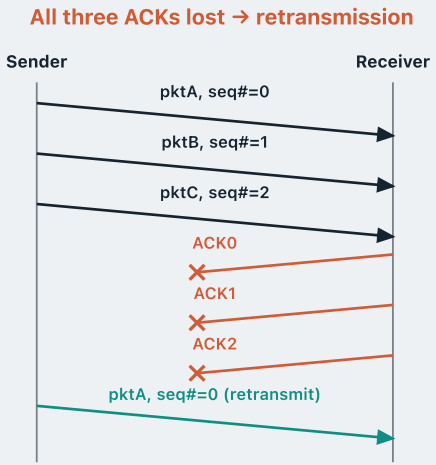
- **Re-send the ACK** — even though this packet was already acknowledged.

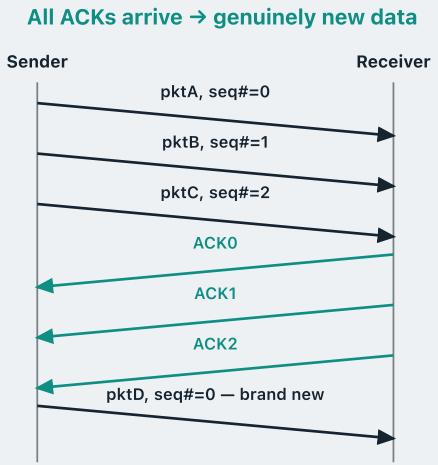
- **Otherwise**: **ignore** the packet.

SELECTIVE REPEAT

A dilemma when window size exceeds the seq# space

- Sequence numbers: **0, 1, 2** — they wrap around: ..., 0, 1, 2, 0, 1, 2, ...
- Window size: **3**. Consider two different histories:



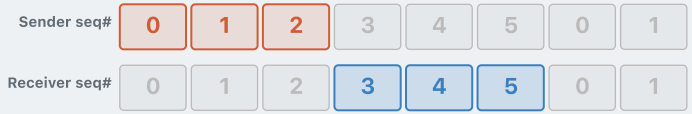


From the sender's side, it knows whether this is a **retransmission** or **genuinely new data**. But from the receiver's perspective, it just sees a packet with **seq#=0** — it cannot tell whether this is a retransmission or genuinely new data. This creates ambiguity.

SELECTIVE REPEAT

Window size vs. seq-number space

The dilemma arises because the sender's window and the receiver's window can drift far enough apart to **overlap after wrap-around**.



THE RULE | window size ≤ ½ × seq# space

e.g. if the window size is **3**, the seq# space should be at least 0–5

Think about it: why is this exactly the right requirement?

RECAP

Module 4 — what we covered

RDT aims to transfer data reliably over an unreliable channel; it is built on **ARQ** (Automatic Repeat reQuest).

- **rdt 2.x** assumes the channel may incur **bit errors**, so it adds checksums, ACKs and additionally, a sequence number distinguishes a retransmission from new data.
- **rdt 3.0** assumes the channel may **lose packets**, so it adds a timer to handle packet loss.

In all those versions we used **stop-and-wait** — sending one packet at a time — which is rather inefficient, so we **pipelined** with a window:

- **Go-back-N** uses cumulative ACKs; if a packet arrives out of order, the sender resends the whole window.
- **Selective repeat** uses receiver buffering and individual ACKs, so the sender resends only what was lost.

References

James F. Kurose & Keith W. Ross, *Computer Networking: a Top-Down Approach Featuring the Internet*, Addison Wesley, 3rd edition, 2005.

Chapter 3 — The Transport Layer (§3.4)

William Stallings, *Data and Computer Communications*, Pearson, 10th edition, 2013.

Chapter 7 — Data Link Control Protocols

Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5th edition, 2010.

Chapter 3 — The Data Link Layer (§3.3, §3.4)

Acknowledgements

These lecture notes are developed based on slides from the following sources:

- **Dr Barry Wu**'s slides for COSC264, University of Canterbury, 2024–25
- **Dr Mengmeng Ge**'s lecture notes for COSC264, University of Canterbury, 2024
- **Assoc Prof Dan Kim**'s lecture notes for COSC264, University of Canterbury, 2017
- **Prof Aleksandar Kuzmanovic**'s lecture notes for CS340, Northwestern University, 2005, https://users.cs.northwestern.edu/~akuzma/classes/CS340-w05/lecture_notes.htm

Lineage & colophon

These lecture notes modernise COSC264 material developed by **Dr Barry Wu** (2024–25), building on notes by **Dr Mengmeng Ge** (2024) and **Assoc Prof Dan Kim** (2017), University of Canterbury, and lecture notes by **Prof Aleksandar Kuzmanovic** (CS340, Northwestern University, 2005).